Probabilistic Emissivity Retrieval From Hyperspectral Data via Physics-Guided Variational Inference

Joshua R. Tempelman, Kevin Mitchell, Adam J. Wachtor, and Eric B. Flynn

***Abstract*—Recent research has proven neural networks to be a powerful tool for performing hyperspectral imaging (HSI) target identification (ID). However, many current methods deliver a single material class prediction for a single spectrum input; such approaches are limited in their interpretability and restricted to predicting materials that are accessible in available training libraries. In this work, we present an inverse modeling approach in the form of a physics-conditioned (PC) generative model. A probabilistic latent-variable model learns the underlying distribution of HSI radiance measurements and produces the conditional distribution of the emissivity spectrum. Moreover, estimates of the HSI scene's atmosphere and background are used as a physically relevant conditioning mechanism to contextualize a given radiance measurement during the encoding and decoding processes. Furthermore, we employ an in-the-loop augmentation scheme and physics-based loss criteria to avoid bias toward a predefined training material set and to encourage the model to learn physically consistent inverse mappings. The generative model may be queried to produce multiple emissivity candidates for a given radiance observation, allowing for uncertainty quantification. Moreover, a distribution-based material matching scheme is presented to return a set of likely material matches for an inferred emissivity distribution. The framework was additionally validated on in situ airborne long-wave infrared (LWIR) hyperspectral data collected over the Los Angeles Basin, where it successfully recovered sulfur-like emissivity features consistent with laboratory references, demonstrating its ability to generalize to real-world observations. Hence, we present a strategy to incorporate contextual information about a given HSI scene, capture the possible variation of underlying material spectra, and provide interpretable probability measures of a candidate material accounting for a given remotely sensed radiance measurement.**

***Index Terms*—Emissivity retrieval, hyperspectral imaging (HSI), remote sensing, variational modeling.**

I. Introduction

LONG-WAVE infrared (LWIR) hyperspectral imaging (HSI) is an effective means for identifying materials from remote (often aerial) sensors thanks to the unique atmospheric and thermal emission properties in the 7–14-μm regime of the electromagnetic spectrum [1], [2]. As such, common application spaces of LWIR-HSI technologies include agricultural production monitoring [3], [4], mineral exploration [5], and urban monitoring [6], [7]. LWIR-HSI frameworks for target material identification (ID) may generally be split into two categories: radiance-based classifiers and emissivity-retrieval models. The former seeks to draw either a categorical or binary decision regarding the ground-based material given the observed radiance [8], [9], while the latter seeks to first estimate the spectral emissivity of the surface, which may be subsequently matched to those of known substances [10], [11], [12], [13]. While material ID models are convenient in the sense that their decision making is largely performed under-the-hood, their interpretation is not straightforward, as they are typically 1) restricted to providing categorical answers with limited clarity regarding uncertainty quantification and 2) can only predict materials that are known a priori to the model, e.g., training materials [14], [15]. Likewise, while numerous atmospheric compensation and temperature-emissivity decoupling techniques exist to perform emissivity retrieval [16], they suffer from the cumulative build-up of uncertainty in solving an ill-posed and over-parameterized inverse problem [17]. Hence, despite the promise of HSI technologies [18], developing interpretable imagining methods still presents multiple challenges due to environmental and atmospheric variation which occlude LWIR-HSI measurements, hence making the task of atmospheric and background estimation a vital component of practical HSI implementation [19].

In recent works, LWIR-HSI frameworks have been supplemented by the introduction of machine learning models which have proven to be well-suited for the task of material ID [20], [21], [22]. Moreover, the flexibility offered by deep-learning architectures and training schemes have allowed for the easy inclusion of multimodal information, such as LiDAR [23], [24], as well as physics-based guidance [15]. As such, many current HSI efforts utilize deep-learning methodologies such as fully connected and graph neural networks to conduct HSI exploitation [25]. However, despite the flexibility of modern deep-learning-based approaches in learning complex recognition tasks, they are generally limited in their interpretability and are commonly applied in a black-box manner.

While recent work has shed light on spectral feature attribution in the context of material classifiers [15], the capacity to account for uncertainty in the over-parameterized inversion of the propagation physics and to provide a candidate set of likely spectral matches to a ground material remains an elusive task. For this reason, we seek a deep-learning-based emissivity-retrieval model, e.g., an *inverse model* that reverses the radiance propagation process and returns the spectra of a target

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The authors are with Los Alamos National Laboratory, Los Alamos, NM 87545, USA (e-mail: tempelman@lanl.gov).

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material in a (possibly) mixed pixel. To robustly configure such a retrieval model, several challenges must be addressed. Namely, stochastically varying atmospheric conditions distort spectral signatures, requiring the compensation of atmospheric effects [17], [26], [27]. In addition, pixels may be composed of multiple mixed materials, with some being of interest and others not, leading to further complications in recovering a target spectrum [28]. Moreover, the inverse problem is ill-posed, making the process of correctly decoupling the effects of atmospheric propagation from the material emissivity highly complicated. For this reason, it's desirable to devise a more flexible modeling framework that is capable of incorporating uncertainty in the inverse process to provide distribution-based estimates of the retrieved material emissivity over the LWIR spectrum.

In this work, we address these challenges by considering a probabilistic inverse model that is conditioned on the outputs maxillary of models, which estimate the scene's radiance propagation parameters, the scene-wide background (nontarget). We focus explicitly on surface emissivity retrieval of solid materials (e.g., gas plumes are not considered). Whereas prior works have investigated probabilistic modeling for estimating the uncertainty of class-label predictions [29] and image segment classification [30], this work is concerned with learning the aleatoric uncertainty of the inverse problem itself. Moreover, the modeling scheme proposed herein focuses on incorporating knowledge of the radiance propagation process. While recent work has demonstrated the auxiliary measurements from an HSI cube are beneficial for contextualizing a pixel of interest, commonly via graph neural networks [31], [32], [33], this work differs by focusing explicitly on learning the radiance propagation properties of HSI scenes to mitigate the over-parameterization of the inverse process. Our model takes the form of a conditioned variational latent-variable model with auxiliary networks that produce conditional atmosphere and background estimates. In this light, our model is *physics-guided*, since it utilizes knowledge of the propagation physics to contextualize the radiance measurements of a given pixel in light of the inversion task.

Accordingly, the remainder of this article is organized as follows. Section II reviews the relevant fundamentals of HSI radiance propagation and provides the mathematical framework for the proposed probabilistic inverse model. Section III outlines the machine learning architecture and information regarding training objectives and learning schemes. The results of several numerical studies are presented in Section IV which highlights the model's performance on hold-out HSI data and showcases its applicability to real-world HSI imagery. Finally, Section V offers concluding remarks and suggestions for future research.

II. Hyperspectral Inverse Problem and Mathematical Framework

We tailor our modeling around the classical radiance propagation equation for LWIR-HSI modeling [2]. This section serves to review its formulation and the classical whitening transformation, motivate the need for a probabilistic modeling framework to learn its inverse, and present a mathematical framework to do so.

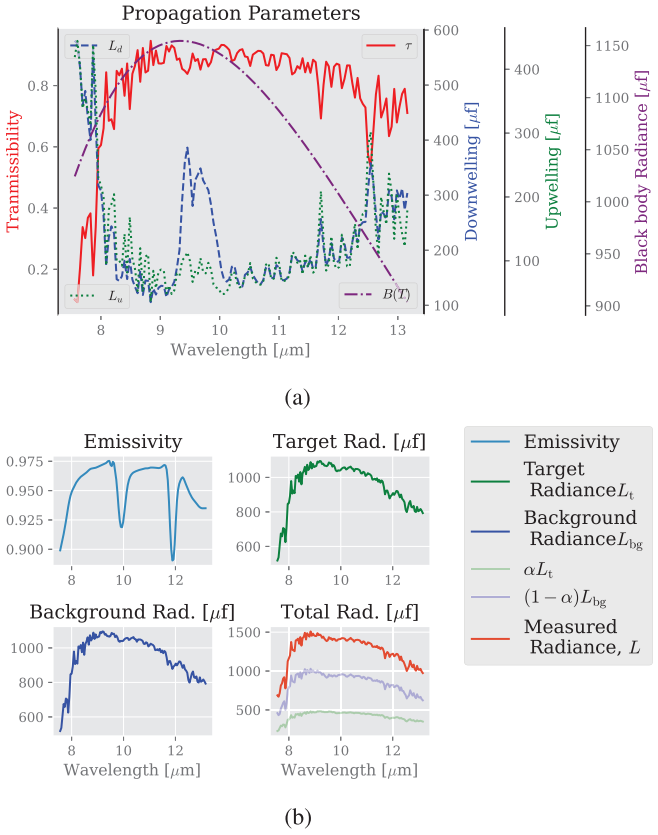


Fig. 1. Example of propagated radiance and atmospheric parameters. (a) Atmospheric parameters $\mathcal{A}$ and (b) target emissivity ϵ, pixel background radiance L_{bg}, target radiance L_t, and propagated radiance L.

A. Radiance Propagation Model and Data Preprocessing

The radiative transfer of thermal emissions to an airborne sensor is a function of the material emissivity ϵ, downwelling radiance L_d, upwelling radiance L_u, transmission τ, and the temperature-dependent blackbody radiance $B(T)$ with T being the average temperature of the scene. In this work, we assume the material reflectance is accurately approximated by $\rho = 1 - \epsilon$. The two primary components accounting for the measured radiance are ground-leaving radiance and ambient radiance. The first is a combination of the ground material emission $\epsilon B(T)$ and the reflected radiance $L_d\rho$, while the latter is comprised of L_u. The atmospheric transmission, τ, determines how much of the ground-leaving radiance reaches the sensor as a function of wavelength [2]. In practice, a material of interest may not fully cover a pixel of an HSI measurement, and hence we consider mixed pixels of target and background radiance of a given pixel (denoted L_{bg}) via the relative strength factor α. Combining target and background radiance, the total at-sensor radiance, $L = f_{\mathrm{p}}(\epsilon; \mathcal{A}, \alpha)$, is calculated as

$$f_{\mathrm{p}}(\epsilon; \mathcal{A}, \alpha) = \alpha \left[\underbrace{\tau(\epsilon B(T) + (1-\epsilon)L_d) + L_u}_{L_t(\epsilon, \mathcal{A})} \right] + (1-\alpha)L_{bg} \quad (1)$$

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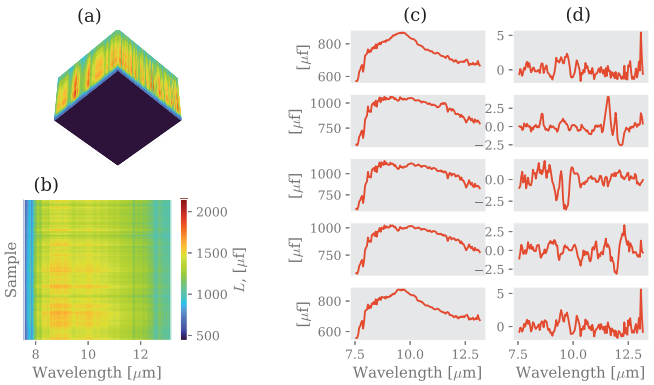


Fig. 2. Cube-whitening process depicting (a) measured hyperspectral data cube C and (b) its reshaped matrix form $\boldsymbol{L} = \mathsf{C}_{(1,2)} \in \mathbb{R}^{(NM)\times r}$ accompanied by specific examples of (c) un-whitened and (d) whitened radiance measures.

where L_t is the radiance propagated from the target material to the sensor and $\mathcal{A} = \{B(T), \tau, L_u, L_d\}$ the set of radiance propagation parameters, e.g., the temperature-dependent Plank function and atmospheric parameters which together are responsible for L_t and L_{bg}. The units for wavelength and radiance are given in microns [μm] and microflicks [μf] unless otherwise stated. Fig. 1 provides a visualization of the various components germane to (1).

Collecting radiance observations over a scene provides a cube of HSI data, $\mathsf{C} \in \mathbb{R}^{N\times M\times r}$ where N, M, and r denote the number of width-wide pixels, height-wise pixels, and the sensor resolution, respectively, where $M = r = 128$ and N varies per cube. Due to the colloquial assumption that $\mathcal{A}$ and L_{bg} are generally consistent throughout a given HSI scene [2], [34], [35], there is typically a strong correlation between radiance measures of a given cube. As such, the matrix form of a cube's radiance measures $\boldsymbol{L} = \mathsf{C}_{(1,2)} \in \mathbb{R}^{(NM)\times r}$ may be decoupled via a whitening transformation

$$\boldsymbol{L}_w = \left(\boldsymbol{L} - \bar{L}_{bg}^{\mathsf{C}}\right)\mathbf{W} \tag{2}$$

where $\bar{L}_{bg}^{\mathsf{C}}$ is the outlier-removed mean background radiance, $\mathbf{W} = \mathbf{V}\boldsymbol{\Lambda}^{-1/2}\mathbf{V}^{\intercal} \in \mathbb{R}^{r\times r}$ is the whitening matrix with $\boldsymbol{\Lambda} \in \mathbb{R}^{r\times r}$ and $\mathbf{V} \in \mathbb{R}^{r\times r}$ being products of the eigendecomposition of the covariance, $\boldsymbol{\Sigma} = \mathbf{V}\boldsymbol{\Lambda}\mathbf{V}^{\intercal}$. In the current work, the covariance is estimated with a subset of $\boldsymbol{L}$, with 20% of the pixels being removed as outliers. Accordingly, for each pixel in the collection C, the set $\underline{L} = \{L, L_w\}$ is returned. Fig. 2 depicts the correlated radiance measures of an example cube C, its reshaped matrix form $\boldsymbol{L}$, and several examples of unwhitened and whitened measurements. Because a majority of the correlation in $\boldsymbol{L}$ is a product of $\mathcal{A}$ and $\bar{L}_{bg}^{\mathsf{C}}$, the whitened signals provide a more expressive representation of the pixel-to-pixel variation which emphasizes specific spectral features of L_t as compared to L alone. In reality, $\bar{L}_{bg}^{\mathsf{C}}$ is not constant throughout the scene, so the pixel-wise background variation of L_{bg} is also present in L_w. Moreover, weakly present signatures (e.g., with a small α value), results in less prominent signatures inL_w .

B. Motivation and Modeling Goal

The task of emissivity retrieval is traditionally performed via atmospheric compensation and temperature-emissivity separation [2]. This is commonly achieved by estimating a scene's atmospheric make-up, i.e., by regressing over a sequence of candidate templates [36] or empirically estimating in-scene atmospheres based on statistical properties of the cube [37]. While doing so enables the estimation of the underlying emissivity of a scene, such approaches are limited in the sense that they cannot accommodate all possible variations in the atmosphere and background, particularly since many software reduce the problem to a low-parameter least-squares matching with predefined atmospheric templates. Moreover, the sought inverse is over-parameterized and ill-posed, rendering it numerically challenging. To this end, recent works have demonstrated that knowledge of the propagation equation, coupled with estimates of the scene's atmospheric composition through numerical optimization programs, allows for the training of *physics-guided* deep learning frameworks which learn to reconstruct the radiance via (1) by inferring all parameters given a single radiance measure [15]. However, the aforementioned framework is tuned to deterministically perform on a subset of defined atmospheric templates and known material spectra, leading to inherent bias and inability to account for uncertainty in the radiance propagation process brought about by un-modeled factors such as turbulence or atmospheric heterogeneity [38], or to account for spectral signals not known to the training library.

For the above reasons, we seek a probabilistic model that, along with estimating an expected emissivity of a given pixel of an HSI scene, incorporates the flexibility of probabilistic modeling and learns the underlying uncertainty of the ill-posed inverse problem. To this end, we seek to learn the probabilistic mapping f based on labeled pairs of data (L, ϵ), with $L \in \mathscr{L}$ being observed radiance $\epsilon \in \mathscr{E}$ the underlying target emissivity, which is also conditioned on estimates of the HSI scenes atmosphere and background (denoted $\hat{\mathsf{C}}$), e.g.,

$$f(L|\hat{\mathsf{C}}) : \mathscr{L} \to \mathscr{E}, f(L|\hat{\mathsf{C}}) : \mathbb{R} \to \mathbb{R} \tag{3}$$

such that wavelength-dependent uncertainty may be captured. To this end, our mathematical objective is to infer the marginal conditional distribution $p(\epsilon|L)$.

C. Mathematical Framework

We begin with the assumption that the emissivity and radiance are related via a joint distribution $p(L, \epsilon, z) : \mathscr{L} \times \mathscr{E} \times \mathscr{Z} \to \mathbb{R}_+$, where $z \in \mathscr{Z}$ is a latent variable. Conditional independence between L and ϵ is assumed through the latent variable z, such that

$$p(L, \epsilon, z) = p(L)p(z|L)p(\epsilon|z) \tag{4}$$

meaning that the shared uncertainty of the inverse mapping may be proxied by an underlying posterior latent distribution that is to be learned. The conditional independence assumption between L and ϵ given z allows the latent space to serve as a manifold where all input-output pairings collapse onto

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structured representations, allowing the model to learn meaningful latent representations while maintaining a probabilistic structure.

The mathematical objective of the proposed framework is to maximize the log-likelihood of $p(\epsilon|L)$

$$\begin{aligned}\max \ell(\epsilon|L) &= \log p(\epsilon|L) \\ &= \log \int p(\epsilon, z|L)\mathrm{d}z. \end{aligned} \tag{5}$$

By (4), this may be equivalently expressed as

$$\ell(\epsilon|L) = \log \int p(z|L)p(\epsilon|z)\mathrm{d}z. \tag{6}$$

Equation (6) is intractable due to the integration over the latent variable. To account for this, a variational approximation of the latent posterior $q(z|L)$ is introduced so that the objective becomes

$$\begin{aligned}\ell(\epsilon|L) &= \log \mathbb{E}_{z\sim q(z|L)}\left[\frac{p(z|L)p(\epsilon|z)}{q(z|L)}\right] \\ &\geq \mathbb{E}_{z\sim q(z|L)}\left[\log \frac{p(z|L)p(\epsilon|z)}{q(z|L)}\right] \end{aligned} \tag{7}$$

with the lower-bound on (7) being a result of Jensen's inequality, thus making it the evidence lower bound (ELBO) of $\ell(\epsilon|L)$ (denoted $\mathrm{ELBO}_{\epsilon|L}$). Applying Bayes' theorem to $p(z|L)$ and factoring out distributions constant under the expectation, the ELBO objective may be written explicitly as

$$\begin{aligned}\mathrm{ELBO}_{\epsilon|L} = \mathbb{E}_{z\sim q(z|L)}&\left[\log p(L|z) + \log p(\epsilon|z)\right] \\ &- D_{\mathrm{KL}}\left[q(z|x)\|p(z)\right] \end{aligned} \tag{8}$$

where $D_{\mathrm{KL}}\left[q(z|L)\|p(z)\right]$ is the Kullback–Leibler (KL) divergence between $q(z|L)$ and $p(z)$, where $p(z) = \mathcal{N}(0, \mathbf{I})$ herein. Equation (8) may be evaluated via amortized inference by considering heteroscedastic Gaussian distributions for $q_\phi(z|x)$, $p_\theta(L|z)$, and $p_\gamma(\epsilon|z)$, where ϕ, θ, and γ are the sought parameters of the assumed model forms. A complete derivation of (8), as well as its numerical implementation, is given in Appendix A.

In addition to the evidence L, we apply a conditioning mechanism in the form of the HSI cube's estimated atmosphere and background (see Section III-D for details). The set of conditioning vectors applied to the input and latent spaces of the generative model are denoted as $\hat{\mathsf{C}} = \{\hat{\mathcal{A}}, \hat{\bar{L}}^{\mathsf{C}}_{bg}\}$ and $c = \{c_{\mathrm{prop}}, c_{\mathrm{bg}}\}$, respectively, so that the ELBO loss which is optimized in the learning process is more completely written as

$$\begin{aligned}\mathrm{ELBO}_{\epsilon|L} = \mathbb{E}_{z\sim q_\phi(z|L)}&\left[\log p_\theta(L|z, c) + \log p_\gamma(\epsilon|z, c)\right] \\ &- D_{\mathrm{KL}}\left[q_\phi(z|x, \hat{\mathsf{C}})\|p(z)\right]. \end{aligned} \tag{9}$$

In this scheme, $\hat{\mathsf{C}}$ estimates the five unknown variables of (1) (excluding the inversion target ϵ), while c serves as a latent representation of $\hat{\mathsf{C}}$.

III. Machine Learning Framework

The machine learning framework follows the conventional supervised learning paradigm. We consider a collection of K HSI cubes recovered from airborne measurements, $[\mathsf{C}] = \{\mathsf{C}_k\}_{k=1}^K$, as well as a collection of M laboratory-measured spectral signatures, $[\epsilon] = \{\epsilon_m\}_{m=1}^M$. For each of the K cubes, the M signatures were injected pixel-wise into the HSI scene per (1) with varying strengths, with the parameters $\mathcal{A}$ coming from conventional physics-based atmospheric estimation routines [1], and the pixel background L_{bg} being taken directly from the measured HSI scene. Further details regarding the particulars of the constructed dataset may be found in Section IV-A. The result is a collection of labeled data pairs (L, ϵ), with accompanying cube parameters $\mathcal{A}$ and $\bar{L}^{\mathsf{C}}_{bg}$, which may be leveraged to learn the distribution $p(\epsilon|L)$.

A. Data Flow and Monte-Carlo (MC) Approximation

We take a generative modeling approach with the goal of estimating $p(\epsilon|L)$ based on a MC estimation. The modeling scheme consists of three main modules which are depicted by Fig. 3 and detailed herein. Namely, a conditioned latent-variable model (also referred to as the generative model or generator) and two complementary models which serve to provide cube-wide conditional information to the generator. The latent-variable model, termed `EpsNet`, is responsible for inferring $p(\epsilon|L)$, i.e., determining probable ϵ given L. The complementary networks, termed `PropNet` and `BgNet`, are responsible for estimating the propagation parameters $\mathcal{A}$ and background radiance $\bar{L}^{\mathsf{C}}_{bg}$, respectively, providing physics-based context to `EpsNet`.

`EpsNet` consists a radiance encoder E_L, a latent normalizing flow model (NFM) f_z, a radiance decoder D_L, an emissivity shape decoder D_ϵ, and an emissivity scale estimator s_{D_ϵ}. The encoder acts on the input set $\{\underline{L}, \hat{\mathcal{A}}, \hat{L}_{\mathrm{bg}}\}$ and produces the vectors μ_ϕ and σ_ϕ to provide latent codes $z \sim \mathcal{N}(\mu_\phi, \sigma_\phi) \in \mathbb{R}^{d_z\times 1}$, with d_z being the latent dimension. The module E_L estimates $q_\phi(z|\underline{L}, \hat{\mathsf{C}})$ which is denoted $q^0_\phi(z|\underline{L}, \hat{\mathsf{C}})$ when sampled prior to the flow model (with corresponding latent codes z or z_0), and $q^K_\phi(z|\underline{L}, \hat{\mathsf{C}})$ after flow operations have been applied (with corresponding latent codes z_K). Likewise, D_L and D_ϵ are parameterized with θ and γ, which learn the parameterizations of $p_\theta(L|z, c)$ and $p_\gamma(\epsilon|z, c)$.

In what follows, x_{E_L}, x_{D_L}, and x_{D_ϵ} denotes data flowing through the j-th layer of E_L, D_L, and D_ϵ, respectively. The encoding blocks compress the input radiance $x^1_{\mathsf{E}_L} \in \mathbb{R}^{128\times 1}$ into a latent representation $x^4_{\mathsf{E}_L} = \{\mu_\phi, \sigma_\phi\} \in \mathbb{R}^{d_z\times 2}$, which defines the mean and standard deviation of the latent distribution. The emissivity decoder D_ϵ performs the inverse operation, expanding $x^1_{\mathsf{D}_\epsilon} = z_k \in \mathbb{R}^{d_z\times 1}$ to $x^4_{\mathsf{D}_\epsilon} = \{\hat{\epsilon}, \sigma^\lambda_{\hat{\epsilon}}\} \in \mathbb{R}^{128\times 2}$. An identical operation is performed by D_L, mapping $x^1_{\mathsf{D}_L} \in \mathbb{R}^{d_z\times 1}$ to $x^4_{\mathsf{D}_L} = \{\hat{L}, \sigma^\lambda_L\} \in \mathbb{R}^{128\times 2}$. D_L and D_ϵ output both the predicted mean and estimated variance of the radiance and emissivity, respectively, and are thus twice the dimension of the spectral length. The mean corresponds to the expected signal (e.g., the predicted emissivity or radiance), while the variance term serves as a learned measure of uncertainty. This variance provides regularization by enforcing reconstruction errors through a negative log-likelihood rather than a simple mean-squared error, encouraging the model to assign higher variance to wavelength regions with greater inherent uncertainty. Between E_L and D_ϵ, the flow model f_z acts on the sampled latent vector

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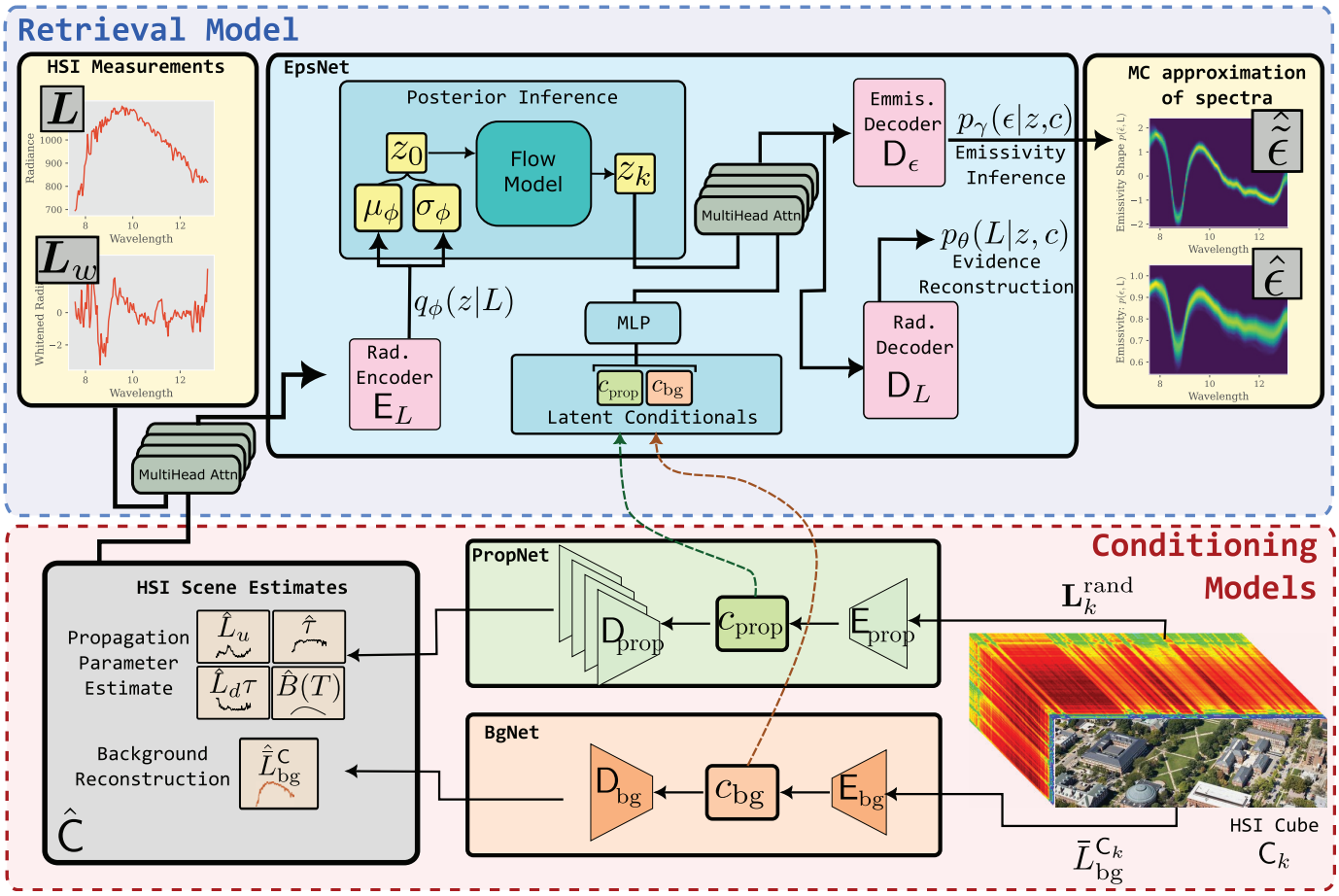


Fig. 3. Data-flow and ML framework. The auxiliary networks PropNet and BgNet estimate the HSI scene parameters for a given cube, while EpsNet estimates $p(\epsilon|L)$ for a given radiance sample from C. CA conditioning between the pixel's radiance and HSI scene estimates applied at the input and latent spaces of EpsNet.

z to produce a transformed distribution with associated sample z_K (as detailed in Section III-B). Altogether, the network's conditional generation of ϵ given L, $\epsilon|L$, is a composition of the functions

$$\hat{\epsilon}|L = \hat{\sigma}_\epsilon \underbrace{\mathsf{D}_\epsilon(f_z(\mathcal{R}(\mathsf{E}_L(\underline{L}, \hat{\mathsf{C}})), c)}_{\hat{\tilde{\epsilon}}|L} + \hat{\epsilon} \tag{10}$$

where $\hat{\tilde{\epsilon}}|L$ is the predicted normalized emissivity shape and $\mathcal{R}$ is the reparameterization of E_L's output, e.g., $\mathcal{R}(\mathsf{E}_L(\underline{L}, \hat{\mathsf{C}}))$ is equivalent to drawing samples $z \sim q_\phi(z|\underline{L}, \hat{\mathsf{C}})$. Finally, to recover the sought empirical distribution, the conditional posterior $q_\phi(z|\underline{L}, \hat{\mathsf{C}})$ is sampled N times and decoded to build the set of inferred emissivity measures, denoted $[\hat{\epsilon}]_N = \{\hat{\epsilon}_n|L\}_{n=1}^N$, leading to the empirical distribution function in normalized and scaled emissivity space

$$\begin{cases} \hat{p}(\tilde{\epsilon}|L) & = \frac{1}{N}\sum_{n=1}^N \delta(\tilde{\epsilon} - \hat{\tilde{\epsilon}}_n(z_n)) \\ \hat{p}(\epsilon|L) & = \frac{1}{N}\sum_{n=1}^N \delta(\epsilon - \hat{\epsilon}_n(z_n)) \end{cases}, \; z_n \sim q_\phi(z|\underline{L}, \hat{\mathsf{C}}) \tag{11}$$

where $\delta(\square)$ is a Dirac measure. We introduce the explicit dependency on z_n in (11) to emphasize the relationship between the variational posterior and the inferred distributions of $\hat{p}(\epsilon|L)$ and $\hat{p}(\tilde{\epsilon}|L)$.

B. Posterior Estimation

To encourage expressivity in the latent distribution while still obeying the KL regularization on $q_\phi^0(z|L)$, we employ a latent NFM. The NFM applies successive invertible mappings to samples drawn from the base distribution $z_0 \sim q_\phi^0(z|L)$ and transforms them to samples of an arbitrarily complex distribution through the composition of mappings, f_k. This transformation of the posterior allows the model to more accurately capture the true posterior distribution, which is certain to be beyond the complexity of the Gaussian prior $p(z)$, and thus encourages more flexible and accurate generative inference.

After the Kth mapping of the normalizing flow, the latent code is

$$z_K = f_K \circ f_{K-1} \circ \cdots \circ f_1(z_0) \tag{12}$$

with corresponding density

$$q_\phi^K(z|L) = q_\phi^0(z_0|L) \prod_{k=1}^K \left|\det \boldsymbol{J}_{f_k}(z_{k-1})\right|^{-1} \tag{13}$$

where $\boldsymbol{J}_{f_k} = \nabla_{z_{k-1}} f_k$ is the Jacobian of the kth mapping. The learnable parameters of f_z are denoted as κ which are updated during training jointly with the other EpsNet modules. We employ the real-NVP flow model in this work [39]. Details

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TABLE I
PARAMETERIZATION OF AUXILIARY NETWORKS

Parameter	Description	Value
d_z	`EpsNet` Latent dimension, $\dim(z)$	64
d_{prop}	`PropNet` Latent dimension	12
d_{bg}	`BgNet` Latent dimension	12
σ_{prop}	`PropNet` Activation	Tanh
σ_{bg}	`BgNet` Activation	Tanh
$d_{h_{\text{prop}}}$	`PropNet` Hidden Dim	128
$d_{h_{\text{bg}}}$	`BgNet` Hidden Dim	128

regarding its implementation and effect on (8) are given in Appendix B.

C. Model Architecture

`PropNet` operates on sets of random radiance samples of a given HSI cube, $\mathbf{L}_k^{\text{rand}} \in \mathbb{R}^{200\times 128}$, with $\mathbf{L}_k^{\text{rand}} \in \mathsf{C}_k \in [\mathsf{C}]$ denoting the kth HSI cube of the dataset. `BgNet` takes in the mean background $\bar{L}_{bg}^{\mathsf{C}_k}$. Both modules encode their inputs, so that the kth cube's latent embedding of the propagation parameters and background are

$$\begin{cases} c_{\text{prop}}^k = m_{\text{prop}}\left(\dfrac{1}{J}\displaystyle\sum_{j=1}^{J}\mathsf{E}_{\text{prop}}(L_j)\right), \; L_j \sim \mathbf{L}_k^{\text{rand}} \\ c_{\text{bg}}^k = \mathsf{E}_{\text{bg}}(\bar{L}_{bg}^{\mathsf{C}_k}). \end{cases} \tag{14}$$

Here, E_{prop} and E_{bg} are five-layer multilayer perceptrons (MLPs) which are tasked with learning the compact representation of $\mathcal{A}$ and L_{bg} of a given cube given raw radiance measurements. m_{prop} is a five-layer MLP that transforms the aggregated output of the deep set encoder into a more informative latent code for subsequent decoding. The dimensions of c_{prop} and c_{bg} is denoted as d_{prop} and d_{bg}, respectively, combing together for a total latent dimension of $d_c = d_{\text{prop}} + d_{bg}$. With this, we have a summary latent representation of a given HSI cube as $c = \{c_{\text{prop}} \oplus c_{\text{bg}}\} \in \mathbb{R}^{d_c}$, where $\oplus$ denotes feature-wise concatenation The networks subsequently decode their respective latent representations to estimate the unknown parameters of $\mathcal{A}$ and the cube's background radiance, respectively, forming the data flows of the kth cube as

$$\begin{aligned} f_{\text{prop}} \quad & \mathbf{L}_k^{\text{rand}} \xrightarrow{\mathsf{E}_{\text{prop}}} c_{\text{prop}}^k \xrightarrow{\mathsf{D}_{\text{prop}}} \hat{\mathcal{A}}_k \\ f_{\text{bg}} \quad & \bar{L}_{bg}^{\mathsf{C}_k} \xrightarrow{\mathsf{E}_{\text{bg}}} c_{\text{bg}}^k \xrightarrow{\mathsf{D}_{\text{bg}}} \hat{\bar{L}}_{bg}^{\mathsf{C}_k}. \end{aligned} \tag{15}$$

D_{prop} represents four independent MLP decoders which map the c_{prop} to the estimate of $\mathcal{A}$, denoted $\hat{\mathcal{A}}$, while D_{bg} is a single MLP decoder which maps the background latent vector back to $\bar{L}_{bg}^{\mathsf{C}_k}$. Altogether, these decoding operations account for the relevant parameters of a given HSI scene, per the assumptions of the adopted propagation model, (1). The parameterization of each auxiliary network is summarized in Table I.

The `EpsNet` encoder and decoders are comprised of four blocks, denoted in general as $Q_j^{\square}$ for the jth block of network $\square$, with each following a Fourier-neural operator (FNO)—MLP architecture depicted in Fig. 4. We denote the input tensor to the jth block of `EpsNet` as x^j. Accordingly, the block-wise operation of encoding and decoding is expressed as the composition

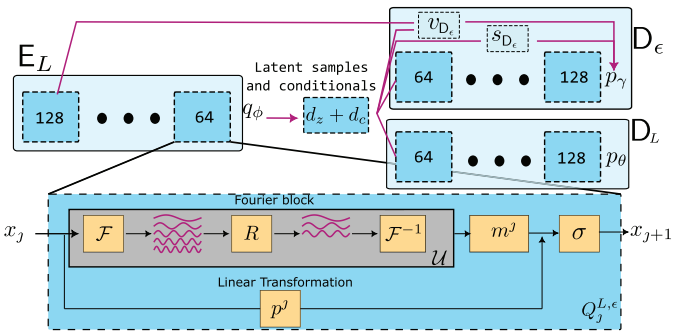


Fig. 4. Encoding and decoding blocks for radiance encoding and emissivity inference. The jth encoding and decoding block consists of FNO-MLP blocks composed of a Fourier convolution operation $\mathcal{U}$, a residual MLP network m^j, a projection layer p^j, and output activation σ.

$$\begin{aligned} \mathsf{E}_L &= Q_1^{\mathsf{E}_L} \circ \cdots \circ Q_N^{\mathsf{E}_L} \\ \mathsf{D}_L &= Q_1^{\mathsf{D}_L} \circ \cdots \circ Q_N^{\mathsf{D}_L} \\ \mathsf{D}_\epsilon &= Q_1^{\mathsf{D}_\epsilon} \circ \cdots \circ Q_N^{\mathsf{D}_\epsilon} \\ Q_j^{\square}(x^j) &= p^j(x^j) + m^j\big(\mathcal{U}\big(x^{j-1}\big)\big). \end{aligned} \tag{16}$$

Here, p^j is a linear projection layer which aligns to the output dimension of x^{j+1} enabling an addition residual operation, m^j is a four-layer residual-net (ResNet) MLP comprised of fully connected layers, and $\mathcal{U}(x^j)$ is a spectral convolution layer

$$\mathcal{U}(x) = \int \varsigma(x,\xi)v(\xi)\mathrm{d}\xi = \mathcal{F}^{-1}\left(R_\chi \mathcal{F}(v)\right)(x) \tag{17}$$

where $\mathcal{F}\{\square\}$ is the Fourier transform of $\square$, and ς is a representative kernel function which is learned in the frequency domain by the linear transformation R_χ operating on a reduced basis of Fourier modes. The primary motivation for employing Fourier layers is that the spectral decomposition captures smooth global representations of the data in an infinite-dimensional function space, enabling the model to better learn compact latent representations for tasks involving functional transformations [40], [41].

The primary decoding blocks of D_ϵ learn the normalized emissivity, $\tilde{\epsilon}$, such that each spectral sample is z-standardized per its own statistic. This is done to avoid bias toward spectra with high amplitude features, as low-variance spectra (which are common in the available library) are difficult to learn from data-driven error metrics alone. For this reason, the normalized emissivity *shape* is predicted separate from its mean and standard deviation, $\hat{\bar{\epsilon}}$ and $\hat{\sigma}_\epsilon$. In this sense, the errors in misrepresenting spectral features are equally applied to each sample irrespective of a material's relative variations within the LWIR bands. An additional module within D_ϵ which is termed `ScaleNet` is tasked with inferring the spectral mean and standard deviation of each sample based on samples from the posterior

$$\{\hat{\bar{\epsilon}}, \hat{\sigma}_\epsilon\} = s_{\mathsf{D}_\epsilon}\big(\{z_K \oplus c\}\big), \quad z_K \sim q_\phi^K(z|\underline{L}, \hat{\mathsf{C}}). \tag{18}$$

The architecture of `ScaleNet` is a five-layer MLP with hyperbolic tangent activation that maps $\{z_K \oplus c\} \in \mathbb{R}^{d_z+d_c}$

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to $\{\hat{\bar{\epsilon}} \oplus \hat{\sigma}_\epsilon\} \in \mathbb{R}^2$, with all hidden layers possessing 128 dimensions. To estimate variance in the LWIR bands, e.g., learn where to allow for greater uncertainty in the inverse mapping by assigning wavelength-dependent leniency in misrepresenting spectral features, an additional network termed `VarianceNet` v_{D_ϵ} operates on the input sequence to E_L and conditioned latent sequence

$$\sigma^2_{\tilde{\epsilon}} = v_{\mathsf{D}_\epsilon}\left(\{x^1_{\mathsf{E}_L} \oplus z_K \oplus c\}\right), \quad z_K \sim q^K_\phi(z|\underline{L}, \hat{\mathsf{C}}). \tag{19}$$

The architecture of `VarianceNet` is a five-layer MLP with each hidden layer possessing 128 dimensions. The parameters of `ScaleNet` and `VarianceNet` are denoted as ϖ. For each MLP submodule in E_L, D_ϵ, and D_L, the hidden dimension matches that of x^j, while the last layer and projection layer compresses or expands the data to the output dimension of x^{j+1}. We utilize self-gated activation in each decoding network (commonly known as Swish activation), and Sigmoid activation in the encoding network. The free network hyperparameters were taken as the latent dimension $d_z = 64$, the number of blocks $n_b = 4$, and the number of MLP layers per block $n_l = 4$; these parameters were determined via Bayesian optimization using a Matern 5/2 kernel and the maximum expected improvement acquisition function [42].

D. Physics-Based Conditioning Mechanisms

The conditional networks serve to provide contextual information to the `EpsNet` based on the propagation model. The goal is to mitigate the ill-posedness and over-parameterization of (1)'s inverse, allowing `EpsNet` to contextualize a single-pixel measurement-based approximations of $\mathcal{A}$ and $\bar{L}^{\mathsf{C}}_{bg}$ provided by the auxiliary network. The conditioning to the input layer of `EpsNet` is applied via a cross-attention (CA) module between the estimated parameters and the observed radiance, while the decoder inputs are conditioned by the CA between the latent codes sampled from the posterior distribution and the latent representation of the cube parameters

$$\begin{aligned} x^1_{\mathsf{E}_L} &= \mathrm{CA}\big(\underline{L}, \{\hat{\mathcal{A}} : \hat{L}_{\mathrm{bg}}\}\big) = \mathrm{CA}\big(\underline{L}, \hat{\mathsf{C}}\big) \\ x^1_{\mathsf{D}_\epsilon}, x^1_{\mathsf{D}_L} &= \mathrm{CA}(z_K, \{c_{\mathrm{prop}} : c_{\mathrm{bg}}\})) = \mathrm{CA}(z_K, c) \end{aligned} \tag{20}$$

where: is channel-wise concatenation. In this scheme, the sensor measurement set $\underline{L}$ and its latent representation serve as the query set that possesses the primary information pertinent to emissivity retrieval, while the estimates of the HSI scene and their respective latent representations serve as the keys which possess contextual information about $\underline{L}$. This allows `EpsNet` to produce conditional posterior estimates that account for the unknown variables in (1), as well as conditioned emissivity inference. In the input attention case, a final mean-aggregation is taken to produce a single vector to `EpsNet` for each observed pixel.

The proposed conditioning framework not only merges auxiliary data streams but also embeds the physical dependencies governing long-wave radiative transfer. Specifically, the estimated transmissivity (τ), upwelling (L_u), downwelling (L_d), and temperature-dependent blackbody radiance $B(T)$ describe how surface-emitted and reflected radiation propagate to the sensor. By conditioning both the encoder and decoder on these quantities, the latent representation is implicitly constrained to respect the physics of (1), while also remaining flexible enough account for model mismatch to real-world complexity. This regularization enables the network to reduce ambiguity between emissivity and atmospheric effects and yields uncertainty patterns that mirror real physical sensitivity—higher variance in low-τ regions and lower variance in transparent windows. Consequently, the physics guidance functions as a radiative-transfer-aware prior that steers the inversion toward physically consistent solutions rather than as a purely empirical adjustment.

We note that the physics-based models used to generate the training data for `PropNet` and `BgNet` may themselves carry uncertainties and modeling errors. Such errors can, in principle, propagate through to the emissivity produced by `EpsNet` which is conditioned on the estimated propagation parameters. However, the variational formulation of our framework mitigates this effect by encoding the conditioning variables as an abstract guide during the probabilistic inversion process, rather than enforcing the retrieval model to strictly respect the estimated propagation parameters. Consequently, the physics-based conditioning mechanism serves to regularize the inverse mapping while maintaining flexibility to accommodate uncertainty and imperfections in both the training data and real-world observations.

E. In-the-Loop Augmentation

To encourage the model to learn a more general inverse mapping, as opposed to recalling a set of M predefined spectral signatures available in $[\epsilon]$, the training and testing emissivity need to be sufficiently diversified. To ensure this, the components of $[\epsilon]$ were modified at each training epoch so that no two target spectra are ever the same during training. Because on-the-fly emissivity generation is currently infeasible with laboratory measurements and intractable via first-principals modeling, a synthetic emissivity manipulation scheme was adopted. Moreover, since the overarching goal of the model is to produce the map $f : \mathscr{L} \to \mathscr{E}$, modifications of ϵ are not strictly required to be constrained to real-world spectra. Nevertheless, the applied perturbations should leave the statistical distribution of the emissivity mean, smoothness, and variance relatively unaffected so that the perturbed spectra emulate realistic spectral shapes and thereby encourage the model to bias toward realistic spectral profiles without memorizing specific spectral of the available training data.

The emissivity perturbation scheme is described here and depicted by Fig. 5. First, the emissivity measures are each z-standardized with respect to their own statistics so as to "normalize" the relative effect of subsequent perturbations, $\tilde{\epsilon} = (\epsilon - \bar{\epsilon})/\sigma_\epsilon$. To ensure sufficient smoothness and spatial structure, the perturbations are applied in the form of a Gaussian process ($\mathcal{GP}$) that is parameterized by an expected value of 0 and a covariance kernel k_{xx} which we take as a Matern 5/2 kernel

$$\tilde{\epsilon}^\star_k = \bar{\epsilon}_k + \mathcal{GP}(0, k_{xx}). \tag{21}$$

Once modified, the emissivity is rescaled per $\epsilon^\star = \mathrm{softclamp}(\tilde{\epsilon}^\star \sigma_\epsilon + \bar{\epsilon})$, where the softclamp operation ensures

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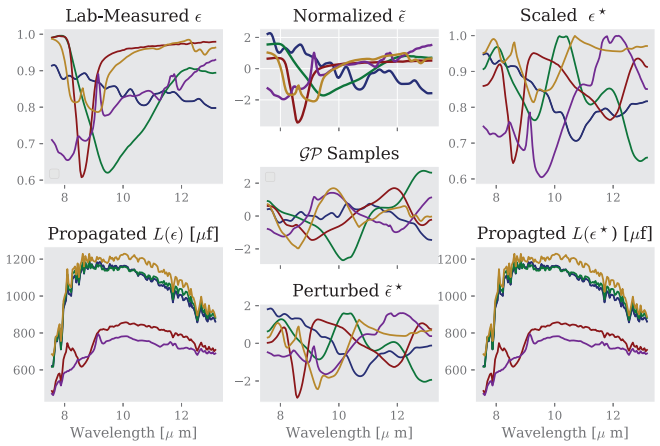


Fig. 5. In-the-loop data augmentation. (Left) Laboratory-measured spectra and their respective radiance measures as taken directly from the training scene in [C]. (Middle) Normalization of the spectral data, samples from a GP, and normalized perturbed spectra. (Right) Scaled perturbed spectra and its repropagated radiance, serving as the *novel* batch sample.

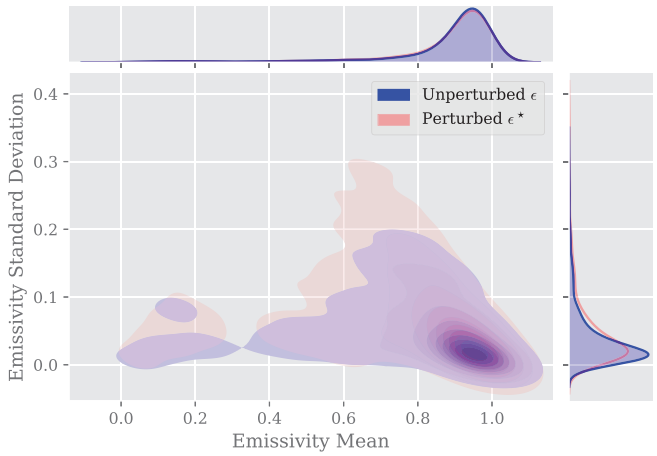


Fig. 6. Comparative statistics of the mean and standard deviations between the unperturbed (lab-measured) spectra, ϵ, and the modulated (perturbed) spectra $\epsilon^\star$. Results are given for a batch sample of size 1024.

the modulated emissivity falls between 0 and 1. The clamped spectra are renormalized as well to provide the set $\{\epsilon^\star, \tilde{\epsilon}^\star\}$. The quantity $\epsilon^\star$ is then repropagated per (1) using the same atmosphere, background, and whitening parameters for the given radiance observation so as to return the propagation of the novel sample into the same scene, $L^\star = f_p(\epsilon^\star; \mathcal{A}, L_{bg})$.

Fig. 5 depicts an example of the emissivity modulation procedure for a mini batch of five pairs of (ϵ, L) (shown in the left column). The $\mathcal{GP}$ is sampled and added to the normalized samples in the center column producing $\hat{\epsilon}^\star$, before being rescaled per the batch's original statistics and repropagated via (1) as shown in the right column. Fig. 6 depicts the preservation of the batch statistics for a batch of 1024 samples, showing the distributions of the mean and standard deviation of emissivity samples before and after modulations. The distributions of ϵ and $\epsilon^\star$ are nearly identical. Moreover, a very high concentration of spectra have nearly unit mean and nearly zero variance, indicating a prominence relatively flat (low-variance) spectral bands, with few high-variance spectra included. After normalization, however, all spectra are statistically equivalent in terms of shape learning via D_ϵ.

It should be noted that the in-the-loop augmentation scheme is not intended as a tunable hyperparameter for performance optimization but rather as a regularization mechanism to prevent the inverse model from overfitting to a finite emissivity library. Within the context of inverse modeling, performing a sensitivity analysis on the perturbation magnitude would not yield physically meaningful trends, since the augmentation does not directly control retrieval fidelity but instead defines the stochastic breadth of emissivity variations used to promote generalization. The augmentation parameters adopted herein were empirically verified to preserve the first- and second-order statistics of emissivity (see Fig. 6), thereby ensuring that the generated perturbations maintain physical realism while encouraging the model to learn robust, physically consistent inverse mappings.

F. Training Scheme

The model training consists of two separate stages: conditional model training and inference model training. The former is responsible for training `PropNet` and `BgNet` in order to first develop models capable of providing the necessary conditioning to `EpsNet`; it is performed offline prior to training `EpsNet`. The latter is responsible for training the learnable parameters of `EpsNet`.

1) Conditional Network Training: To train `PropNet`, 2000 HSI cubes were collected into a dataset with corresponding propagation parameters to form the training dataset $([\mathcal{A}]_{\texttt{PropNet}}, [\mathsf{C}]_{\texttt{PropNet}})$. From $[\mathsf{C}]_{\texttt{PropNet}}$, 100 000 radiance sets (with a single set here denoting $\mathbf{L}_k^{\text{rand}} \in \mathbb{R}^{200\times 128}$) were drawn to form training and validation datasets. A joint training objective was adopted. The first is standard mse loss between $\hat{\mathcal{A}}$ and $\mathcal{A}$, while the second is a propagation objective whereby a flat emissivity curve of $\epsilon = 0.5$ with unit strength is used to propagate the data per (1)

$$\mathcal{L}_{\text{prop}}(\varrho) = \frac{1}{K}\sum_{1}^{k}\Big[(1-\beta)\|\mathcal{A}-\hat{\mathcal{A}}\|_2^2 + \beta\|f_p(0.5,\mathcal{A},1) - f_p(0.5,\hat{\mathcal{A}},1)\|_2^2\Big]. \tag{22}$$

The first loss metric provides a direct enforcement of the atmospheric reconstruction. The latter loss provides a physically meaningful measure with respect to the end goal of emissivity retrieval, as it encourages `PropNet` to learn a parameter set ϱ and latent representation that provides the closest match to true atmospheric composition in terms of the forward propagation process. We note that the selected value of $\epsilon = 0.5$ is lower than average for most materials (see Fig. 6). However, this was selected during training to prevent biasing `PropNet` toward de-emphasizing accurate inference of L_d relative to $B(T)$ [see (1)]. Training is scheduled by the relative weight β which scales linearly from 0 to 1 throughout training over 1000 epochs with a learning rate of 9×10^{-4}.

To train `BgNet`, the set of 2000 training cubes and their outlier-removed means were utilized, $([L_{bg}]_{\texttt{PropNet}},$

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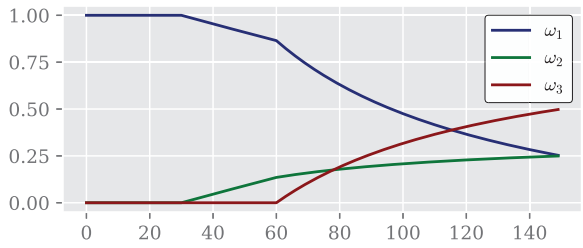


Fig. 7. Relative weights over-training epochs.

$[\mathsf{C}]_{\texttt{PropNet}}$). A standard mse error was applied during training to learn the `BgNet` parameter set υ over 1000 epochs with a learning rate of 10^{-3}.

2) `EpsNet` Training: The process of learning the inverse map, e.g., inferring $p(\epsilon|L)$, is guided by $\mathcal{L}_{\text{inversion}}$ which comprises six different components aimed at i) learning the shape of the normalized spectra and enforcing a measure of smoothness (which is motivated by the continuity requirement for real emissivity curves); ii) learning the relative scaling of the sample, e.g., it's mean and standard deviation; and iii) enforcing a direct error metric before and after scaling the spectra per the mean and standard deviation estimates. In the normalized emissivity space, the error is computed as the negative log-likelihood of the heteroscedastic Gaussian, with the wavelength-dependent variance and mean predicted by D_ϵ. Altogether, the composite loss function is given as

$$\mathcal{L}_{\tilde{\epsilon}} = \underbrace{\mathcal{L}_{\text{shape}} + \mathcal{L}_{\text{smooth}}}_{(i)} + \underbrace{\mathcal{L}_{\text{sdev}} + \mathcal{L}_{\text{mean}}}_{(ii)} + \underbrace{\mathcal{L}_{\tilde{\epsilon}} + \mathcal{L}_{\epsilon}}_{(iii)}. \tag{23}$$

For a batch size of K with spectral length J, the specific form of each subloss is expressed as

$$\mathcal{L}_{\text{shape}} = \frac{\omega_{\text{shape}}}{K} \sum_{k=1}^{K} \frac{\tilde{\epsilon}_k \cdot \hat{\tilde{\epsilon}}_k}{\|\tilde{\epsilon}_k\| \|\hat{\tilde{\epsilon}}_k\|} \tag{24}$$

$$\mathcal{L}_{\text{smooth}} = \frac{\omega_{\text{smooth}}}{K(J-2)} \sum_{k=1}^{K} \sum_{j=0}^{J-3} (\epsilon_{k,j+2} - 2\epsilon_{k,j+1} + \epsilon_{k,j})^2 \tag{25}$$

$$\mathcal{L}_{\text{sdev}} = \frac{\omega_{\text{sdev}}}{K} \sum_{k=1}^{K} \left|\left|\hat{\sigma}_{\epsilon_k} - \sigma_{\epsilon_k}\right|\right|_2^2 \tag{26}$$

$$\mathcal{L}_{\text{mean}} = \frac{\omega_{\text{mean}}}{K} \sum_{k=1}^{K} \left|\left|\hat{\mu}_{\epsilon_k} - \bar{\epsilon}_k\right|\right|_2^2 \tag{27}$$

$$\mathcal{L}_{\tilde{\epsilon}} = \frac{\omega_{\tilde{\epsilon}}}{K} \sum_{k=1}^{K} \sum_{j=1}^{J} \left[\log\left(\sigma^{\lambda}_{\tilde{\epsilon},j}\right)_k^2 + \frac{\left(\hat{\tilde{\epsilon}}_{k,j} - \tilde{\epsilon}_{k,j}\right)^2}{\left(\sigma^{\lambda}_{\tilde{\epsilon},j}\right)_k^2} \right] \tag{28}$$

$$\mathcal{L}_{\epsilon} = \frac{\omega_{\epsilon}}{K} \sum_{k=1}^{K} \|\hat{\epsilon}_k - \epsilon_k\|_2^2 \tag{29}$$

where $\omega_\square$ is a relative scaling term for $\mathcal{L}_\square$. Moreover, by propagating the scaled emissivity predictions back into radiance space per (1) (with the ground-truth measures of $\mathcal{A}$ and $\bar{L}_{bg}^{\mathsf{C}}$ begin assumed available during training), we introduce a propagation loss $\mathcal{L}_{\text{propagation}}$ which provides a measure of how consistent a predicted spectra is with the observed radiance. For a batch of J unique cubes and with K_j radiance examples drawn from the jth cube, the loss is calculated as

$$\mathcal{L}_{\text{propagation}} = \frac{1}{KJ} \sum_{j=1}^{J} \sum_{k=1}^{K_j} \left\| f_{\text{p}}(\hat{\epsilon}_{k,j}; \mathcal{A}_j, \alpha_k) - L_{k,j} \right\|_2. \tag{30}$$

Finally, we enforce the regularization loss, e.g., the KL-divergence objective of the ELBO loss, which is modified to account for the effect of the normalizing flow on the estimated posterior

$$\mathcal{L}_{\text{regularization}} = D_{\text{KL}}\left(q_\phi(z|L)\|p(z)\right) - \sum_k \mathbb{E}_{z_0 \sim q_\phi^0(z_0|L)} \log|\det J_k(z_k)|. \tag{31}$$

Altogether, the joint training objective of `EpsNet` is expressed as

$$\mathcal{L}(\boldsymbol{\vartheta}, \boldsymbol{\omega}) = \omega_1 \mathcal{L}_{\text{inversion}} + \omega_2 \mathcal{L}_{\text{propagation}} + \omega_3 \mathcal{L}_{\text{regularization}} \tag{32}$$

where $\boldsymbol{\vartheta} = \{\theta, \phi, \gamma, \kappa, \varpi\}$ denotes the set of parameters for the various submodules of `EpsNet` which are jointly updated per the common objective (32). During training, we enforce a weight schedule to ω_1, ω_2, and ω_3 which is depicted by Fig. 7. The schedule encourages the model to first emphasize the inversion map in a deterministic fashion (e.g., with regularization and propagation losses ignored). Next, the weight of the propagation loss is slowly introduced, so that the model is encouraged to tailor the inverse predictions to be as physically consistent as possible with the observed radiance. Finally, the regularization is activated to enforce finite latent variance leading to the sought probabilistic mapping.

The `EpsNet` modules were jointly optimized using a learning rate of 10^{-3} for 150 epochs with an exponentiation weight decay coefficient of 0.99 and a weight-regularization constant of 5×10^{-5}. We utilized 2 500 000 radiance samples during training by injecting spectral signatures from a lab-measured library into 2000 HSI scenes, leading to approximately 1200 samples per cube. Further details of the dataset design are given in Section IV-A. The modulation routine described in Section III-E was applied at every epoch with $\mathcal{GP}$ samples generated with unit variance.

IV. Numerical Experiments and Results

To evaluate the proposed modeling scheme, we consider a series of numerical experiments to investigate the model's effectiveness for probabilistic emissivity retrieval. Moreover, we compare the proposed model to an equivalent one that is not privy to physics-based conditioning, e.g., without access to the HSI scene estimates and CA scheme described in Section III-D. In what follows, we describe the datasets used for training and testing (see Section IV-A), overview the learning performances of both models (see Section IV-B), evaluate their accuracy in a variance-aware fashion (see Section IV-C), and demonstrate the practical use of the model as a probabilistic material ID tool (see Section IV-D).

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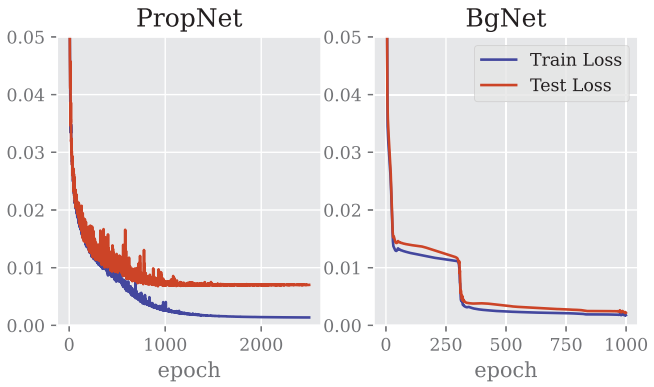


Fig. 8. Loss curves of (left) PropNet and (right) EpsNet training.

A. Dataset Design

Two datasets of hyperspectral data were constructed to train, validate, and test the model over. The measurements of each scene were collected with a Mako airborne sensor as described in [16], with measurement dates spanning between 2018 and 2024. The sensor resolution of each measurement is fixed at 128 wavelength bands between 7.56 and 13.16 μm. Having access to a library of available HSI cube measurements spanning a diverse set of geographical locations, dates, and times, we partitioned a training library of $K_{\text{train}} = 2000$ HSI cubes along with their estimated atmospheric variables $\mathcal{A}$ and background $\bar{L}_{bg}^{\mathsf{C}}$ per conventional physics-based atmospheric compensation. The training library was constructed to only include data collected on even-valued days. To ensure that the training and validation data did not overlap with hold-out (test) data, a separate library of HSI data was constructed consisting of $K_{\text{test}} = 500$ cubes with measurements made on odd-valued dates. In each library, a proprietary set of $M = 10\,095$ laboratory measured LWIR emissivity curves for solid materials was injected into the scene, with the parameters $\mathcal{A}$ propagating the target radiance [L_t of (1)] for the pixel, and with the pixel's measured background radiance accounting forL_{bg} .

We numerically explored the performance of the trained model by generating MC-approximations of $p(\epsilon|L)$ per (11). For a given radiance observation, an inference set size of 1280 samples (10$\times$ the dimensionality) was generated to construct $\hat{p}(\epsilon|L)$, which is sufficient for constructing well-approximated covariance matrices of the empirical distribution. The numerical evaluation of the model was computed for test data that was not introduced to the model during training, e.g., the hold-out set. The target strengths ranged between $\alpha = 0.1$ and $\alpha = 1.0$ for training and testing sets unless otherwise stated.

Although the airborne HSI data used in this work were collected under nominally fixed acquisition geometries, the training corpus spans more than one thousand distinct scenes acquired across different flight campaigns, altitudes, and view angles. As such, while geometry-dependent parameters (e.g., sensor height and off-nadir angle) are not explicitly labeled or modeled, their natural variability is implicitly represented within the training distribution. Consequently, the learned propagation estimates from PropNet and BgNet capture a range of realistic atmospheric and geometric conditions. Moreover, it is important to note that the synthesized dataset used in this work focuses on target-on-background configurations, in which a single laboratory-measured solid spectrum is injected into real HSI scenes. This design isolates the inverse modeling performance of the emissivity-retrieval framework but does not account for multimaterial or subpixel mixing within the target region. Such mixture effects are common in operational HSI sensing and will be explicitly explored in future extensions of this framework.

B. Learning Performance

The training and validation losses of the auxiliary networks (PropNet and BgNet) and primary network (EpsNet) are depicted in Figs. 8 and 9, respectively. Starting with PropNet, we note that over-training begins near the 1000 epoch mark. However, this occurs after sufficient learning of the atmospheric makeup, as is confirmed by the reconstruction examples presented in Appendix C. Hence, despite the seemingly disparate numerical performance of Fig. 8, PropNet sufficiently captures the dominant features of propagation components, resulting in meaningful conditioning information for the modules of EpsNet. Over-fitting is less apparent in BgNet training, and this is due to the relative simplicity of BgNet's learning objective of auto-encoding a 128-D radiance vector. In summary, the training history indicates a meaningful embedding of c_{bg} was learned for the HSI scene background.

The learning results of Fig. 9 are the more germane for the overall modeling goal, with inversion, propagation, and regularization objectives being highlighted separately for a holistic evaluation of the objective (32). For each plot, the training loss, validation loss (e.g., in-distribution hold-outs), testing loss (out-of-distribution hold-outs), and testing loss without physics conditioning are shown. The left six plots depict the six itemized loss functions of $\mathcal{L}_{\text{inversion}}$. During the initial 60 epochs, before regularization is activated, the loss curves for $\mathcal{L}_{\text{propagation}}$ and each component of $\mathcal{L}_{\text{inversion}}$ fall significantly (with the exception of smoothing loss) for both conditioned and unconditioned models, while the reconstruction-based metrics (e.g., $\mathcal{L}_{\text{mean}}$, $\mathcal{L}_{\text{sdev}}$, and $\mathcal{L}_{\epsilon}$) decay more rapidly for the former. During this burn-in stage, the networks learn a pseudo-deterministic inverse map and there is no tradeoff between signal reconstruction and KL-regularization. Beyond epoch 60, KL-regularization is activated per the learning scheme of (32), $\mathcal{L}_{\text{regularization}}$ declines rapidly to a value near 0.5, indicating a satisfactory agreement with the normal prior and hence variability in the mappings. Notably, the inversion and reconstruction losses remain relatively flat for the physics-conditioned (PC) model after the regularization enforcement, indicating that the model retains a majority of its capacity for performing the mapping even under the latent uncertainty. Since millions of examples are evaluated at each epoch, the reconstruction losses of the regularized model are indicative of the error between the target emissivity measures and the model's expectation under sufficiently large sampling. In contrast, these same loss quantities grow for the unconditioned model upon regularization enforcement, indicating that the

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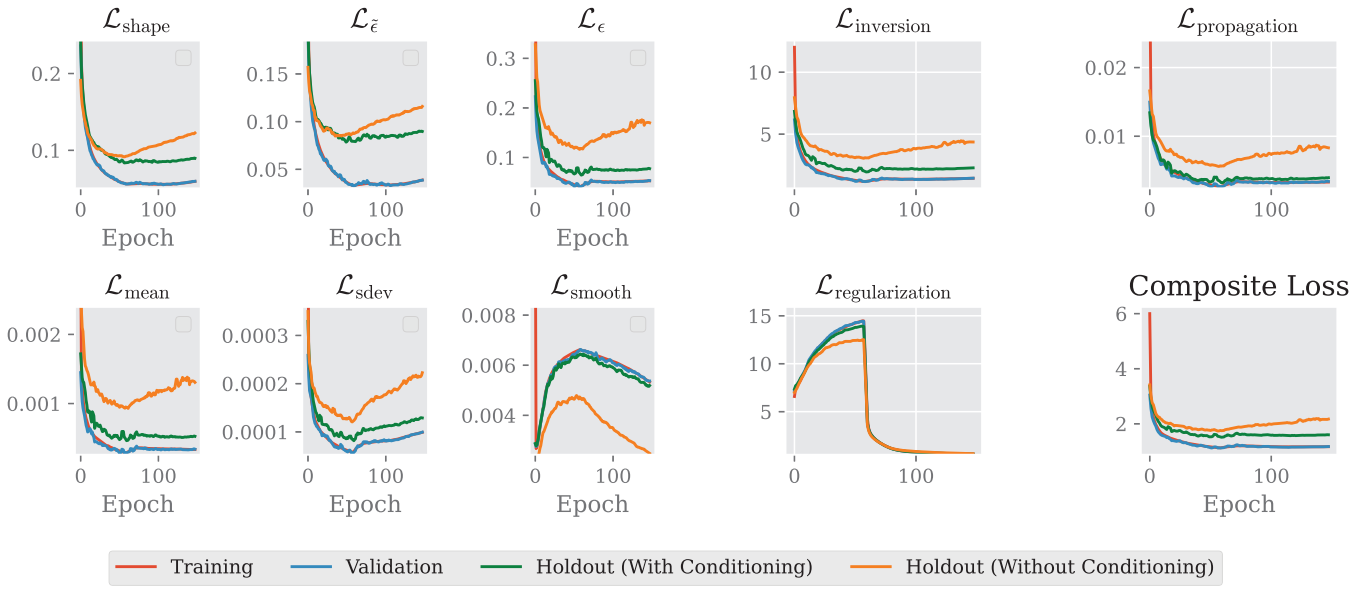


Fig. 9. Loss curves for the training objective defined by (32) depicting training, validation, and testing data, as well as the test loss of the model not privy to conditional data streams. The left three columns depict the itemized components of $\mathcal{L}_{\text{inversion}}$, whereas the right two columns depict the three over-arching objectives and their normalized combination.

unconditioned model struggles to simultaneously satisfy the reconstruction and regularization objectives.

From the above results, we deduce the following: 1) that the whitened data allows for the relatively accurate learning of spectral shape (with or without physics conditioning); 2) that conditional information $\hat{\mathbf{C}}$ is highly potent for contextualizing the relative magnitude of the L_t given an observed radiance measure; and 3) that the physics contextualization is important for the model to jointly satisfy both the inversion and regularization objectives. Briefly expanding on 2), we note that the amplitude of L_t is a function of both the temperature $B(T)$ and relative pixel fill α. Hence, by conditioning on the HSI scene, the model implicitly learns the relative prominence of L_t within L by contextualizing the observation with the estimated blackbody function and atmospheric parameters of each cube.

C. Statistical Evaluation

We now seek to evaluate the empirical distributions generated by MC-sampling per (11). We begin with a qualitative evaluation of the generated empirical emissivity distributions with kernel density estimates (KDEs) of MC samples. Fig. 10 depicts the resulting KDE of scaled emissivity, $\text{KDE}(\{\epsilon_n|L\}_{n=1}^N)$, for 18 randomly selected test samples. The "ground truth" emissivity, e.g., the emissivity which was propagated through (1) to generate the input radiance measure, is superimposed with a thick black line. For the majority of samples shown, this line falls well-within the high-density regions of the KDE indicating a good match to the inferred distribution. The expected value of the MC approximation (denoted $\mathbb{E}_{z\sim q_\phi}[\epsilon|L]$) and the 2-$\sigma$ variance bounds are also plotted, with good agreement being shown between the expected value of the inferred distribution and target emissivity which accounted for the input radiance measurement. Hence, Fig. 10 demonstrates (in a qualitative sense) that the goal of estimating a conditional distribution $\hat{p}(\epsilon|L)$ carrying finite uncertainty is achieved by the model.

We next seek to quantify the match of the inferred distribution to the target emissivity numerically. To do so, we consider a log-likelihood metric of the ground-truth emissivity ϵ being a member of the inferred distribution $\hat{p}(\epsilon|L)$. Taking the empirical mean and standard deviation of the MC distribution as $\mu_{\hat{p}}(\lambda)$ and $\sigma_{\hat{p}}(\lambda)$, where the argument λ explicitly notes wavelength dependence, the likelihood for a given emissivity spectrum being apart of the model's inferred distribution can be readily computed over each wavelength band of the LWIR spectrum. The results of the likelihood estimation are summarized by Fig. 11 for both normalized emissivity ($\tilde{\epsilon}$) and scaled emissivity (ϵ), and for both PC and unconditioned modeling schemes. The solid lines and shaded regions of Fig. 11 denote the average likelihood and standard deviation of the likelihood for 500 random samples of test data.

Both models display comparable performance with respect to emissivity shape, with the log-likelihood ranging between -1.5 (error of $\lesssim 2$ standard deviations) and 0.5 (error of $\lesssim 1$ standard deviations) indicating a very strong match between the target spectra and inferred distribution in general. There is a notable wavelength dependence on the likelihood scores, with the models struggling to infer the correct shape in the low-wavelength regions (between 7.5 and 9 μm). The increased ambiguity is a result of the low transmission in these bands, occluding most ground-leaving radiance for the low-wavelength regions (see Fig. 1). This finding suggests there is indeed a correlation between the physics of remote HSI and the interpretable uncertainty produced by the variational workflow. Moreover, the comparable performance between conditioned and unconditioned models reaffirms the empirically drawn conclusion that whitened radiance possesses rich information regarding the spectral features of the underlying

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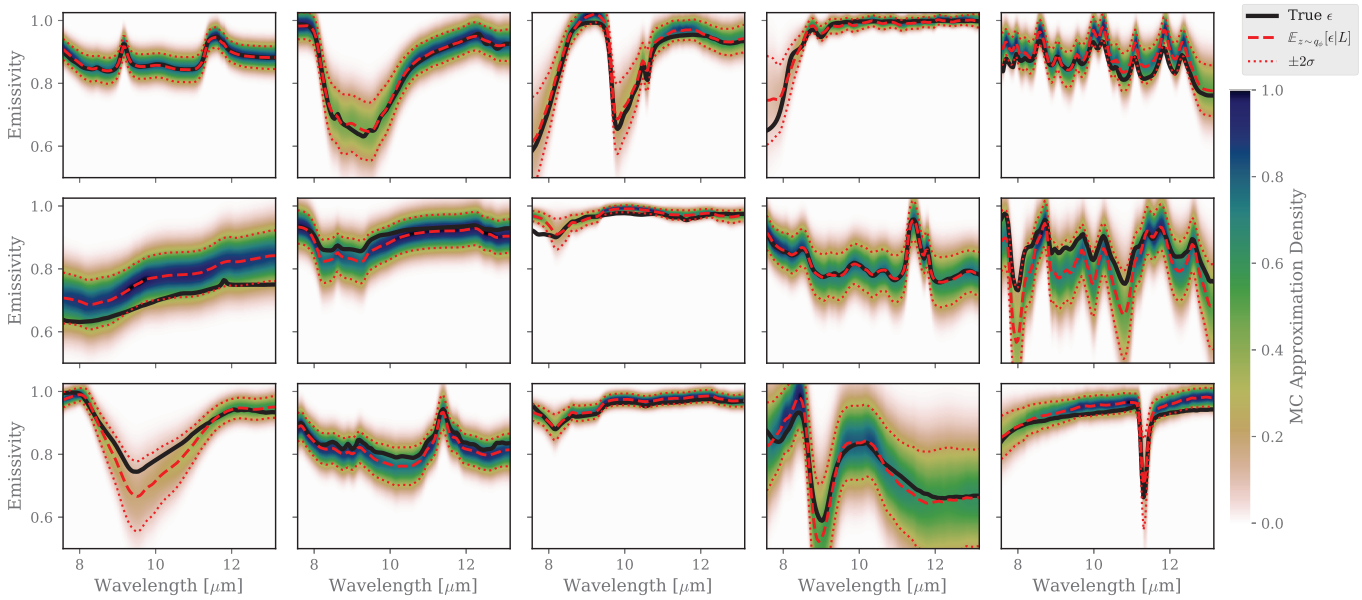


Fig. 10. Visualizations of $\hat{p}(\epsilon|L)$. The color map depicts the KDEs for sampled sets of emissivity of size 1280, while the thick and thin dashed-red lines depict the empirically estimated expectation and 2σ variance threshold, respectively. The ground truth (e.g., the emissivity which was propagated into the observed radiance measure) is depicted with a thick black line.

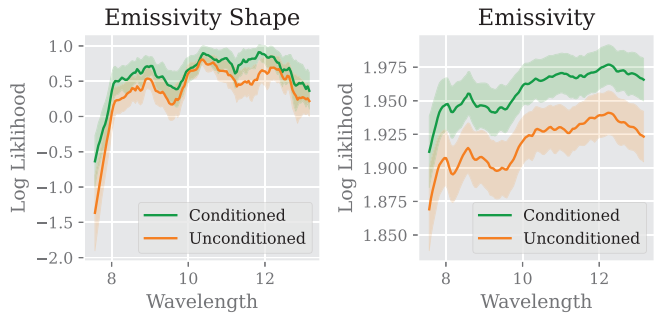


Fig. 11. Wavelength-dependent likelihood estimates for the inferred emissivity shape and scaled emissivity for both the PC and unconditioned models. Results are shown for a dataset of 500 radiance, with 1280 MC-samples comprising $\hat{p}(\epsilon|L)$ for each sample.

emissivity, with physics-based conditioning offering some (but not significant) improvement. We note this also aligns with the loss curves presented in Fig. 9.

Regarding the scaled emissivity of Fig. 11, higher log-likelihood scores and greater deviation between the conditioned and unconditioned modeling schemes are observed on average. Moreover, there is less wavelength-dependent variation, which is expected due to the prominence of relatively low-variance emissivity curves. For such signals, capturing the correct mean (or offset) of the emissivity curve is vital; incorporating the physics-based conditioning modules is thus highly beneficial for learning the relative amplitude and variance of the target emissivity curve for a given radiance sample. This too aligns with Fig. 9, where the PC model outperforms the unconditioned model for tasks such as scaled emissivity reconstruction and propagation reconstruction. Hence, while both models show good performance in terms of capturing the target spectra via $\hat{p}(\epsilon|L)$, a performance increase is recovered via physics-based conditioning.

D. Material ID

With the end goal of most HSI modeling frameworks being the remote detection of materials, we conduct a final set of numerical experiments to evaluate the efficacy for material ID. To do so, the laboratory measured materials $[\epsilon]$ were propagated into test HSI cubes via (1) without applying $\mathcal{GP}$ modulations. The generated empirical distributions were subsequently compared to the candidate spectra to determine the most-likely candidates. In what follows, the matching performance in a qualitative sense for a set of four common materials is presented first, and a quantitative analysis of the likelihood of returning the target material as a top-K match for various values of K is subsequently given.

1) Variance-Aware Framework for Material Matching: The spectral matching process is as follows. After producing $[\hat{\epsilon}]_N$ and $[\hat{\tilde{\epsilon}}]_N$ for the propagated radiance measures, the most-likely materials may be identified by quantifying how well each spectrum of $[\epsilon]$ aligns with the empirical distribution. To draw such comparisons, we consider the Mahalanobis distance between the empirically generated set of N spectral curves, $[\hat{\epsilon}]_N$, and each candidate spectra $\epsilon \in [\epsilon]$, which we denote as $d_{\text{MD}}(\hat{p}, \epsilon)$. Defining $\boldsymbol{\mu}_{\hat{p}}$ and $\boldsymbol{\Sigma}_{\hat{p}}$ and the mean vector and covariance matrix of $[\hat{\epsilon}]_N$, respectively, the measure $d_{\text{MD}}(\hat{p}, \epsilon)$ is given as

$$d_{\text{MD}}(\hat{p}, \epsilon) = \sqrt{(\epsilon - \boldsymbol{\mu}_{\hat{p}})^{\intercal} \boldsymbol{\Sigma}_{\hat{p}}^{-1} (\epsilon - \boldsymbol{\mu}_{\hat{p}})}. \tag{33}$$

With this definition, the match-score between an empirically generated distribution of normalized and scaled emissivity (denoted $\tilde{\hat{p}}$ and $\hat{p}$ herein) and a corresponding scaled and unscaled emissivity candidate from the spectral library is given

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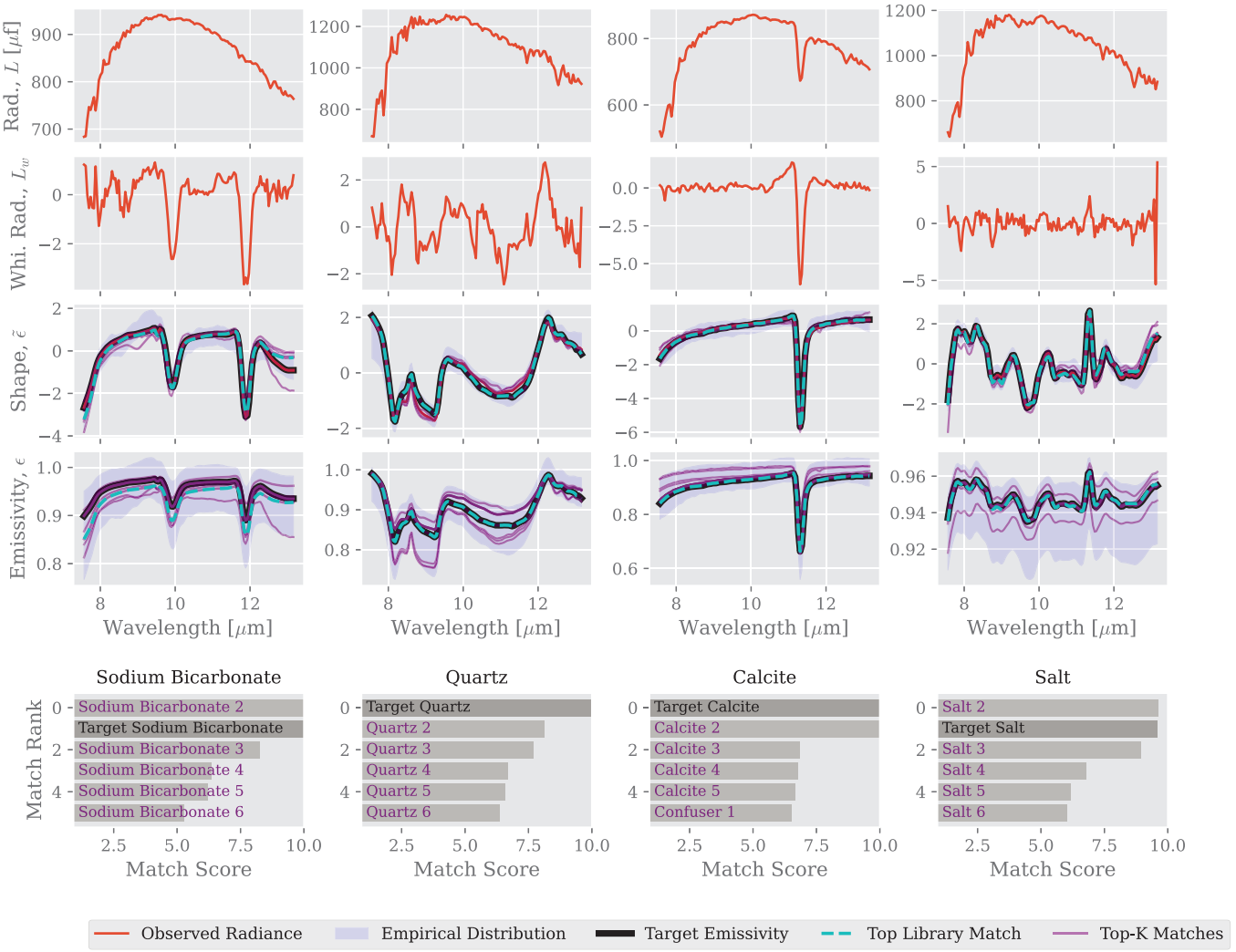


Fig. 12. Probabilistic material ID. (Top) Observed radiance values L and whitened radiance L_w for four sample materials propagated into an HSI scene. (Middle) Distribution of the estimated emissivity shape $\hat{p}(\tilde{\epsilon}|L)$ and scaled emissivity ϵ in shaded blue with the true emissivity in black, top match in dashed cyan, and top-K matches in purple. Bottom: abbreviated material names from the library of the top-K matches along with their match score, with the true target material appearing in black.

as a combined measure of d_{MD} applied in both the normalized (shape-matching) and scaled (overall spectra matching) spaces

$$M(\hat{p},\epsilon) = \left[\underbrace{\left(d_{\text{MD}}(\tilde{\hat{p}},\tilde{\epsilon}) + d_{\text{MD}}(\hat{p},\epsilon)\right)^{-1}}_{M_0(\hat{p},\epsilon)} - \zeta \right]^{1/2} \tag{34}$$

where the quantity $\zeta = \{\min M_0(\hat{p},\epsilon); \epsilon \in [\epsilon]\}$ is a normalization constant used so that 0 is the least-likely score of the library. A key advantage of this metric is its ability to account for variance dependencies over wavelength thanks to the incorporation of the full covariance matrix, thus making it a variance-aware distribution-based metric.

The probabilistic material ID framework was applied to the propagated radiance of measured samples of baking soda, quartz, calcite, and salt. These materials were selected for three primary reasons. First, they are common materials with relative prominence in everyday HSI scenes. Second, for each, there exists multiple similar measurements in the library (typically of the same material, but with different parameters, e.g., grain size and color) allowing us to measure the model's ability of returning high-probability scores for materials with similar spectra. Finally, the selected materials exhibit diversity in spectral features across the LWIR spectrum allowing for the demonstration of the model on four unique material classes.

The results of the material ID experiment are given by Fig. 12. The empirically estimated distributions $\hat{p}(\epsilon|L)$ and $\hat{p}(\tilde{\epsilon}|L)$ (depicted with blue shading) are well-aligned with the spectral signature of the target material for each example. The top match (shown in dashed cyan) and ground truth (shown in black) are the same for two of the four examples, indicating the highest score of (34) being designated to the correct material. The top-6 matches are shown for each example, along with their scores from (34). For the samples where the top match did not align with the sample material, the identified top match is similar to the target spectrum in shape and scale. Moreover, the top material predictions of Fig. 12 are all of the same material class (i.e., quartz and salt) as the target

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spectrum. The results show that the target material is returned as a top-5 match for each example, and that high probability is assigned to the target material even if it is not recovered as the top match. Moreover, when looking to the alternative materials that are selected as highly probable, it is seen that they are all similar in nature to the target material, typically having the same composition but with varying measurement hyperparameters (such as particle size).

The results of Fig. 12 highlight several notable advantages of the probabilistic inverse modeling framework. Whereas traditional classification-based target ID networks return a discrete label without a clear explanation of confidence or likely alternatives, the presented model herein correctly returns high-probability spectra and a measure of alignment for each material. Moreover, for the examples of Fig. 12, the nearly equally high scores are returned for materials of the same material class. This affirms the model's ability to model general spectral features of the inverse process in a regularized (probabilistic) framework, thus encouraging the mapping of more general material classes rather than fixating on data-specific features (as is common when training deterministic models on a fixed training set). This introduced flexibility is a result of the enforced uncertainty of the latent distribution and the emissivity modulation scheme described in Section III-E. Hence, we conclude that the developed probabilistic inverse model also provides a meaningful tool for probabilistic material ID.

2) Quantitative Comparison to Variance-Neglecting Frameworks: As a last experiment, we evaluate the "hit rate" of the model as a function of the number of candidate spectra considered, denoted K. In this context, a "hit" is defined as the model returning the target material within the top K matches to the inferred emissivity distribution. Moreover, we consider a pseudo-deterministic limit of our model by comparing the matching scheme of (34) to two expectation-based measures. Denoting the expected value for $\hat{p}(\epsilon|L)$ over samples of the latent posterior as $\mathbb{E}_{z\sim q_\phi}\left[\hat{p}(\epsilon|L)\right]$, we can consider distance metrics to the empirical expectation to be a variance-neglecting matching scheme. To this end, we consider the cosine distance (CD) and L_2 norm as matching criteria between elements of the candidate spectra set $[\epsilon]$ and the expectation vectors

$$d_{\mathrm{CD}}(\hat{p},\epsilon) = \mathrm{CD}\left(\mathbb{E}_{z\sim q_\phi}\left[\hat{p}(\epsilon|L)\right],\epsilon\right) \tag{35}$$

$$d_{\mathrm{L2}}(\hat{p},\epsilon) = \left|\left|\mathbb{E}_{z\sim q_\phi}\left[\hat{p}(\epsilon|L)\right]-\epsilon\right|\right|_2. \tag{36}$$

Since the wavelength-dependent variability and variation correlations that are present in $\hat{p}(\epsilon|L)$ are not captured by $\mathbb{E}_{z\sim q_\phi}\left[\hat{p}(\epsilon|L)\right]$, the expectation-based criteria is considered variance-neglecting. Comparing the hit rate of the variance-aware (34) to the expectation-based (35) and (36) provides a direct measure to test the benefit of the probabilistic modeling framework in the context of material ID.

The results of the hit-rate experiment are depicted by Fig. 13 for subtrial datasets with minimum strengths of $\alpha = 0.1, 0.25, 0.5$, and 0.75, and for $K \in [1, 1000]$. Curves showing hit-rate versus K are given for both conditioned and unconditioned modeling schemes, and for the three proposed material matching frameworks. Regarding the interpretation of the curves, the faster the convergence to 1, the better the performance. In this light, the proposed variance-aware and PC matching framework is the clear front-runner for all subtrials. Ranking second is the nonconditioned (NC) and variance-aware framework, which, along with its conditioned counterpart, outperforms all pseudo-deterministic matching frameworks; this indicates the benefit of incorporating structured uncertainty when identifying candidate materials. Moreover, the performances of all schemes improve as the minimum allowable strength grows which is an expected result as more of the target emissivity is present in the signal (as opposed to background) for higher α values.

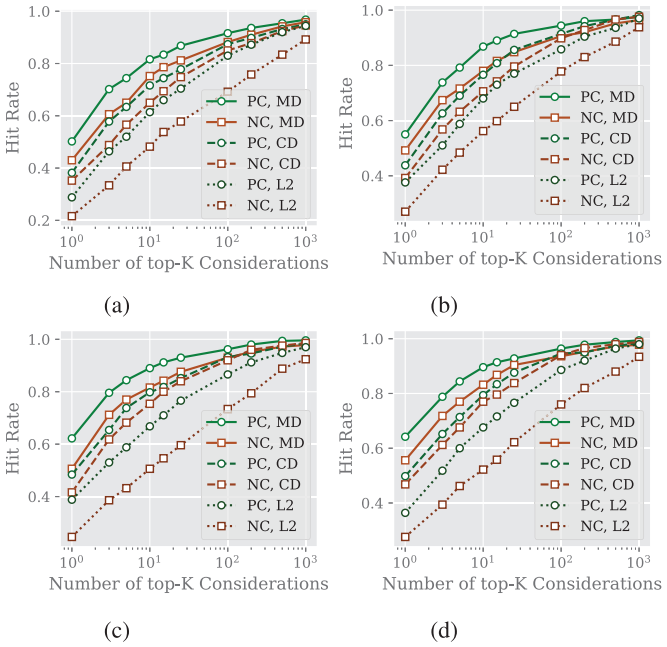


Fig. 13. Probability of including the target material spectrum in the top-K matches (e.g., hit rate) for sample sets with minimum allowable strengths of (a) 0.1, (b) 0.25, (c) 0.5, and (d) 0.75. The six lines correspond to PC versus NC models, and to distribution-based (MD), expectation-based CD, and expectation-based L2 distance (L2) matching schemes.

Commenting on the overall performance of the PC and variance-aware results of Fig. 13, it is shown that even for datasets containing low-α samples that the probability of identifying the specific target measurement is over 80% when incorporating just the first ten matches. Moreover, per Fig. 12, it is likely that the other nine matches are highly similar to the target in spectral makeup. When weak signals are excluded (e.g., $\alpha > 0.75$), this metric raises near 90%. Hence, the results of Fig. 13 lead to the following conclusions: 1) that the probabilistic material ID framework is effective for return target materials with high success rates; 2) that the physics conditioning results in a quantitatively more effective inverse model; and 3) that awareness of the structured uncertainty over the wavelength domain is beneficial for performing robust matching with the inverse model, e.g., that the probabilistic nature of the model is beneficial for material ID. Moreover, shown in Fig. 13, the PC framework consistently outperforms the NC variant, with the largest improvement observed at low target strengths ($\alpha \leq 0.25$), thereby demonstrating the

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necessity and benefit of the physics guidance provided by PropNet and BgNet.

The improvement observed in Fig. 13 under low-signal conditions ($\alpha \approx 0.25$) provides evidence regarding which aspects of the physics conditioning most strongly enhance retrieval. The auxiliary estimates of τ and $B(T)$ from PropNet effectively normalize radiance amplitude across scenes with differing atmospheric loads or surface temperatures, allowing the inverse model to better isolate emissivity shape rather than thermal contrast. Inclusion of L_u and L_d information further helps the network suppress spurious correlations arising from atmospheric self-emission or reflection. These physically informed cues narrow the feasible emissivity space and improve probabilistic ranking of candidate materials, particularly when target radiance is weak or background variability is large. Thus, the observed performance gain for the PC model arises directly from embedding radiative-transfer awareness into the inference process.

To verify that this improvement stems from the physical relevance of the conditioning variables rather than architectural complexity, a partial ablation was performed in which the outputs of PropNet and BgNet were randomly permuted across HSI cubes prior to inference. The results are given in Appendix D. Under these randomized conditionals, the model received atmospheric and background parameters unrelated to the actual scene. In this configuration, the hit-rate advantage of the PC model disappeared—its performance converged to that of the unconditioned case but did not decline further. This outcome indicates that the retrieval framework leverages the conditioning information only when it carries physically consistent context, confirming that the gains reported in Fig. 13 originate from meaningful radiative-transfer guidance rather than from architectural bias.

E. Demonstration on In Situ Data

To further verify the findings presented in Fig. 12, which were demonstrated using synthesized test data, we conducted an initial evaluation of the proposed probabilistic retrieval framework on real, in situ hyperspectral imagery. This assessment serves as a preliminary validation step to confirm that the physics-guided inverse model can recover meaningful emissivity signatures from genuine airborne observations. Evaluating the model on real-world collections introduces several inherent challenges. First, verifying retrieval accuracy requires knowledge of the true ground-leaving emissivity of the target material—information that is rarely available for open-source or archived HSI datasets. Second, although the synthetic experiments in this study realistically simulate radiative transfer and atmospheric propagation, they neglect subpixel mixing effects that are unavoidable in practical airborne measurements. Consequently, while the present analysis focuses on single-material retrieval, the influence of spectral mixing remains an open topic for future work.

For this evaluation, we utilized airborne LWIR-HSI data collected over the Los Angeles Basin, as described in [16]. Within this dataset, a region containing sulfur deposits was identified using a traditional, physics-based material-ID code developed in-house. Pixels corresponding to this sulfur-rich area—where the conventional algorithm indicated high confidence of target presence—were then reanalyzed using the proposed ML-based retrieval framework. A view of the localized region hosting the sulfur match is shown in Fig. 14(a). The region of interest

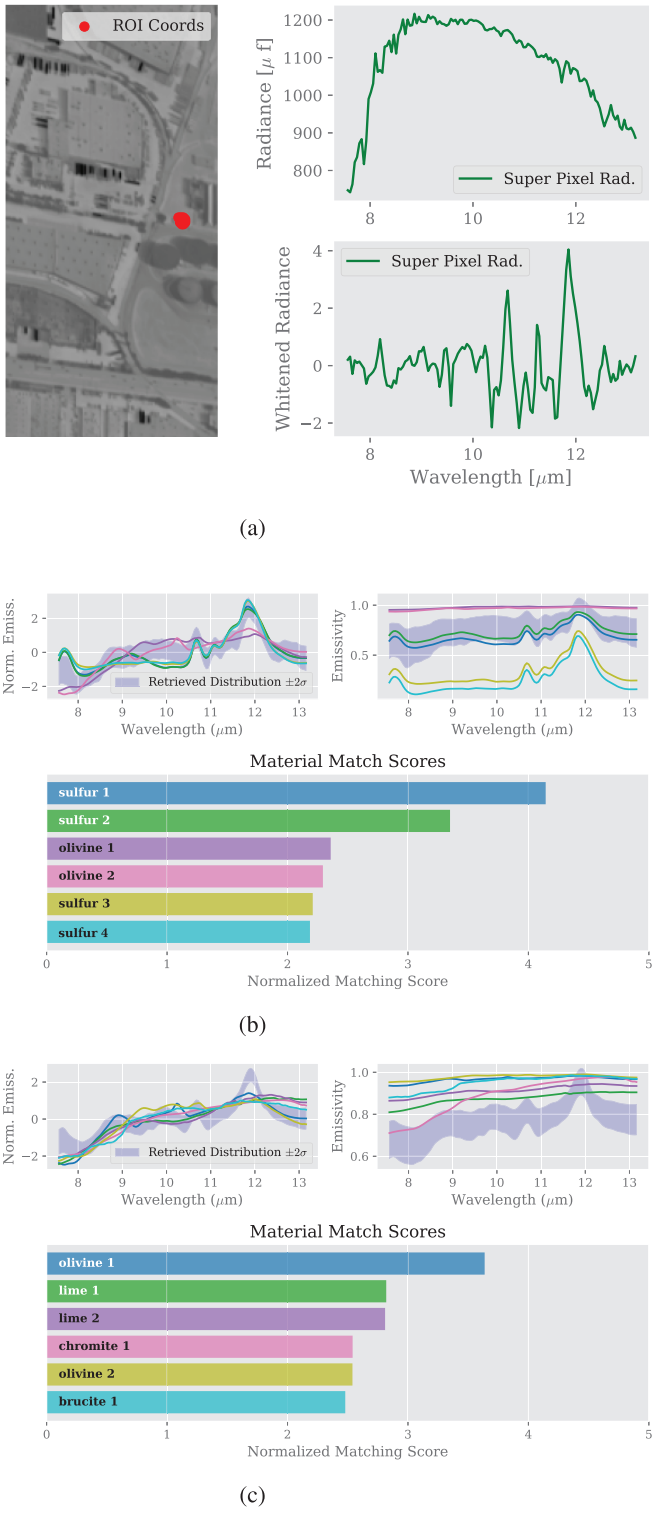


Fig. 14. In situ data collection over the LA Basin showing (a) sulfur ROI and corresponding super-pixel radiance, along with retrieval and material matching results from the ML inversion model for (b) unconditioned and (c) PC frameworks.

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(ROI), flagged for elevated sulfur concentration, is indicated by red dots, with the corresponding super-pixel unwhitened and whitened radiance spectra shown in green. The surrounding facility is an Ecovyst sulfuric acid processing plant, providing additional contextual support for the presence of solid sulfur. Although ground-truth emissivity measurements were not available, we are confident in the model's ID given the corroborating industrial context. The coordinates of the ROI are 33°50'40.8"N, 118°13'50.7"W.

Both the PC and unconditioned versions of the model were applied directly to the same ROI pixels. The conditional networks estimated the scene's propagation parameters and background encoding over a neighborhood of 256×128 256×128 pixels surrounding the ROI. The resulting emissivity distributions—both normalized and unnormalized—predicted by the conditioned and unconditioned models are shown in Fig. 14(b) and (c), respectively. In each subfigure, the upper panels display the retrieved emissivity distributions (in light blue), overlaid with the top six matches from an in-house library of laboratory-measured solid materials. Because available sulfur references vary primarily in brightness while maintaining similar normalized spectral shapes, we increased the weighting of the normalized Mahalanobis distance when performing material matching. The corresponding bar plots show the identified materials and their associated match scores.

Despite not being retrained or fine-tuned on real data, the inferred emissivity distributions recovered distinct sulfur-like spectral features consistent with laboratory sulfur references. Furthermore, the PC framework produced higher-confidence matches to the presumed ground-truth material (sulfur) than the unconditioned model. Although these results are preliminary, they demonstrate the feasibility of applying the proposed probabilistic retrieval framework to in situ airborne data without additional retraining. A more comprehensive validation—including quantitative comparison against field-verified emissivity measurements and an investigation of mixed-material effects—will be the focus of future work.

V. Conclusion and Suggested Future Work

This work presented a deep-learning framework for probabilistic emissivity retrieval, e.g., a hyperspectral inverse model with quantifiable uncertainty. The conditional distribution $p(\epsilon|L)$ was inferred based on whitened and un-whitened sensor measurements, for which an ELBO objective was derived. A latent-variable model was constructed to satisfy the training objective which utilized a combination of ResNet and FNO-MLP type encoding and decoding operations, with a normalizing flow applied to the latent posterior. Furthermore, a physics-based conditioning mechanism was incorporated whereby the auxiliary parameters of the propagation equation dictating the mapping from ϵ to L [e.g., $\mathcal{A}$ and L_{bg} of (1)] were learned based on surrounding cube's radiance measurements, providing vital contextualization to a single-pixel's observation for the task of inverting (1). The efficacy of the proposed modeling strategy was demonstrated through several numerical experiments. It was shown that the MC estimates of $p(\epsilon|L)$ effectively capture target emissivity, and that the physics-based conditioning improves inference of $p(\epsilon|L)$ as compared to unconditioned modeling (see Figs. 9–11). Moreover, we demonstrated that the proposed model may perform as a probabilistic material ID mechanism (see Fig. 12), and that a PC and probabilistic framework for material matching outperforms unconditioned and pseudo-deterministic schemes (see Fig. 13). In addition to the synthetic evaluations, the model was applied to in situ LWIR hyperspectral data collected over the Los Angeles Basin. In this real-world test, the framework successfully recovered emissivity distributions consistent with known sulfur spectral signatures, even without retraining on field data. This demonstration highlights the model's capacity to generalize beyond controlled, synthetic conditions and to yield physically meaningful retrievals from genuine airborne observations.

The aforementioned findings lead to the general conclusion that the proposed model is effective for performing targeting emissivity retrieval with quantified uncertainty. With this comes several notable advantages and implications. First, the proposed framework provides a means for both quantifying the likely spectral variability in the LWIR range given a radiance measure and for determining likely material matches (in contrast to standard categorical classification models). Second, the proposed deep learning framework is relatively cheap to query (in contrast to conventional atmospheric compensation codes [1]). Moreover, its application extends beyond training distributions which was verified by separating testing and training cubes by acquisition date (a challenge for deep learning frameworks); this generality is thanks in part to the physics-based conditioning mechanisms. Finally, in contrast to conventional emissivity-retrieval algorithms which consider all ground-leaving radiance of a pixel (e.g., mixed background and target), our proposed framework learns to decouple the background emissivity from the target spectrum (e.g., the portion of the pixel's emissivity that is not considered background), even when the relative target strength is low (on the order of 10 Hence, this work provides valuable assets to analysts which can be leveraged in LWIR HSI sensing and exploitation efforts. These real-world findings reinforce the broader applicability of the approach and demonstrate that the proposed probabilistic inverse model can be used as a practical retrieval and material-ID tool for operational HSI analysis.

While this work has focused on synthesized data, e.g., data which was generated by propagating laboratory measured spectra (or perturbations of it) into measured HSI scenes via (1), future work could further verify this modeling framework on real-world measurements (in contrast to the current work which considers laboratory-measured spectra synthesized into real-world HSI scenes). Of course, such a task is accompanied by many challenges, such as knowing the ground-leaving target emissivity and having the resources to perform the accompanying aerial measurements. Future work could also consider more intricate conditioning schemes, i.e., by overriding the uniform background and atmosphere assumption in lieu of localized conditioning (in contrast to cube-wide conditioning) for a given ROI. Finally, as this work focused exclusively on solids, future work could extend this framework to gas plume ID.

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APPENDIX A
EVALUATION OF THE ELBO LOSS

Here, we provide the complete details of the derivation and numerical implementation of (8) [or equivalently (9)], along with details regarding its implementation.

A. Derivation

We begin with (6) and introduce a variational distribution $q_\phi(z|L)$ to make the integration tractable

$$\begin{aligned}\ell(\epsilon|L) &= \log \int \frac{q(z|L)}{q(z|L)} p(z|L)p(\epsilon|z)\mathrm{d}z \\ &= \log \mathbb{E}_{z\sim q(z|L)}\left[\frac{p(z|L)p(\epsilon|z)}{q(z|L)}\right] \\ &\geq \mathbb{E}_{z\sim q(z|L)}\left[\log \frac{p(z|L)p(\epsilon|z)}{q(z|L)}\right] \end{aligned} \tag{37}$$

which is the un-factorized ELBO objective. We may expand (37) as

$$\mathrm{ELBO}_{(\epsilon|L)} = \mathbb{E}_{z\sim q(z|L)}\left[\log p(z|L) + \log p(\epsilon|z) - \log q(z|L)\right]. \tag{38}$$

Noting $p(z|L) = p(L|z)p(z)/p(L)$ by Bayes theorem, this turns to

$$\log p(z|L) = \log p(L|z) + \log p(z) - \log p(L) \tag{39}$$

thus furnishing the objective

$$\begin{aligned}&\mathrm{ELBO}_{(\epsilon|L)} \\ &= \mathbb{E}_{z\sim q(z|L)}\Big[\log p(L|z) + \log p(z) \\ &\qquad - \underbrace{\log p(L)}_{\text{constant}} + \log p(\epsilon|z) - \log q(z|L)\Big]. \end{aligned} \tag{40}$$

Since $p(L)$ is constant within the expectation computed over samples of $z \sim q(z|L)$, it may be omitted from the objective giving

$$\begin{aligned}&\mathrm{ELBO}_{(\epsilon|L)} \\ &= \mathbb{E}_{z\sim q(z|L)}\Big[\log p(L|z) + \log p(\epsilon|z)\Big] \\ &\qquad + \underbrace{\mathbb{E}_{z\sim q(z|L)}\Big[\log p(z) - \log q(z|L)\Big]}_{-D_{\mathrm{KL}}(q(z|L)\|p(z))} \end{aligned} \tag{41}$$

which is the final form reported by (8).

B. Implementation

We utilize amortized inference and assume variational model forms

$$\begin{aligned} q_\phi(z|L) &\coloneqq \mathcal{N}(\mu_\phi, \sigma_\phi^2),\ \{\mu_\phi, \sigma_\phi^2\} = f_\phi(L) \\ p_\theta(L|z) &\coloneqq \mathcal{N}(\mu_\theta, \sigma_\theta^2),\ \{\mu_\theta, \sigma_\theta^2\} = f_\theta(L) \\ p_\gamma(\epsilon|z) &\coloneqq \mathcal{N}(\mu_\gamma, \sigma_\gamma^2),\ \{\mu_\gamma, \sigma_\gamma^2\} = f_\gamma(L). \end{aligned} \tag{42}$$

We assume diagonal (but not constant) variances for all distributions, e.g., heteroscedastic Gaussian. With this, the evaluation of (8) may be numerically implemented as

$$\mathbb{E}_{z\sim q_\phi(z|L)} \log p_\theta(L|z)$$

$$= \mathbb{E}_{z\sim q_\phi(z|x)}\left[\log \frac{1}{\sqrt{2\pi\sigma_\theta^2}} - \frac{\|L^i - \mu_\theta\|^2}{2\sigma_\theta^2}\right] \tag{43}$$

$$\mathbb{E}_{z\sim q_\phi(z|L)} \log p_\gamma(\epsilon|z)$$

$$= \mathbb{E}_{z\sim q_\phi(z|x)}\left[\log \frac{1}{\sqrt{2\pi\sigma_\gamma^2}} - \frac{\|\epsilon^i - \mu_\gamma\|^2}{2\sigma_\gamma^2}\right] \tag{44}$$

$$D_{\mathrm{KL}}\left(q(z|L)\|p(z)\right) = -\frac{1}{2}\left(1 - \log\sigma_\phi^2 - \mu_\phi^2 - \sigma_\phi^2\right). \tag{45}$$

APPENDIX B
NORMALIZING FLOW

The normalizing flow applied a series of invertible functions to a sample of the base distribution $z_0 \sim q_\phi^0(z_0|L)$. In this work, the realNVP flow model presented in [39] is applied based on its scalability, expressivity, and efficient Jacobian calculations in comparison to competing flow models. Briefly summarizing its structure, the flow model operates on samples of the base distribution by partitioning the latent sample z_0 into z_{0a} and z_{0b}, and applying then computing their updated values as

$$\begin{aligned} z_{1b} &= z_{0a} \\ z_{1b} &= z_{0b} \odot \exp(s^{(1)}(z_{0a})) + t^{(1)}(z_{0a}) \end{aligned} \tag{46}$$

where $s^{(k)}(\square)$ and $t^{(k)}(\square)$ are scale and translation functions which are implemented as trainable MLP networks. For each successive layer of the flow, the partitioning order is switched so that the first or second halves of z_k is designated as z_{ka} and z_{kb} in alternating fashion. The RealNVP is invertible with a block-diagonal Jacobian

$$\frac{\partial f_k}{\partial z_{k-1}} = \begin{bmatrix} \mathbf{I} & \mathbf{0} \\ \frac{\partial z_{kb}}{\partial z_{(k-1)a}} & \mathrm{diag}(\exp(s(z_{(k-1)a}))) \end{bmatrix}. \tag{47}$$

Thanks to the block-diagonal structure, the transformed density of the posterior may be conveniently expressed as

$$\log q_\phi^K(z|L) = \log q_\phi^0(z_0|L) - \sum_{k=1}^{K}\sum_{i=1}^{N_k} s_i^{(k)}(z_{ka}) \tag{48}$$

where N_k is the length of $s^{(k)}$'s output. Accordingly, the KL-divergence of the transformed posterior takes the form

$$\begin{aligned} D_{\mathrm{KL}}[q^k(z|L)|p(z)] &= D_{\mathrm{KL}}[q_\phi^0(z_0|L), p(L)] \\ &\quad - \mathbb{E}_{z_0\sim q_\phi^0(z_0|x)} \sum_{k=1}^{K}\sum_{i=1}^{N_k} s_i^{(k)}(z_{ka}). \end{aligned} \tag{49}$$

APPENDIX C
ATMOSPHERE RECONSTRUCTION

Here we provide brief details of the reconstruction capacity for `PropNet` given samples from an HSI scene. Fig. 15 depicts the atmospheric parameters estimated from conventional physics-based regression codes as well as those inferred from `PropNet`. Each row of Fig. 15 depicts the results for a different HSI cube. We note a good general agreement between the two for the three atmospheres considered herein, and comment that a similar level of agreement is achieved by all cubes in general. The specific cubes and atmospheric selected for display by Fig. 15 were chosen randomly.

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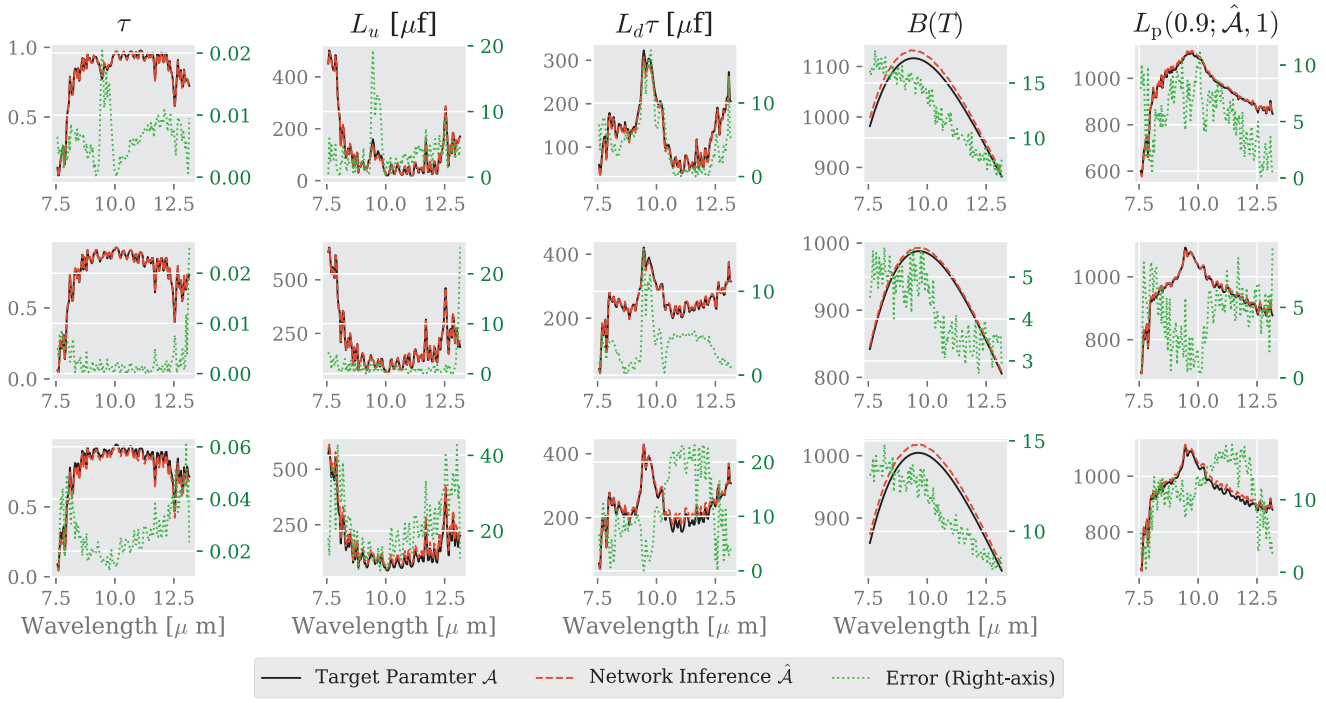


Fig. 15. Propagation parameter estimates for three HSI cubes. Each row depicts a different cube, while the columns depict the transmission τ, upwelling L_u, downwelling-transmission product $L_d\tau$, and blackbody radiance B. The far-right column depicts the result of (1) for true and predicted parameters for a constant emissivity of 0.9 and unit strength. The error for each component is depicted in green and corresponds to the right axis of each subplot in scale.

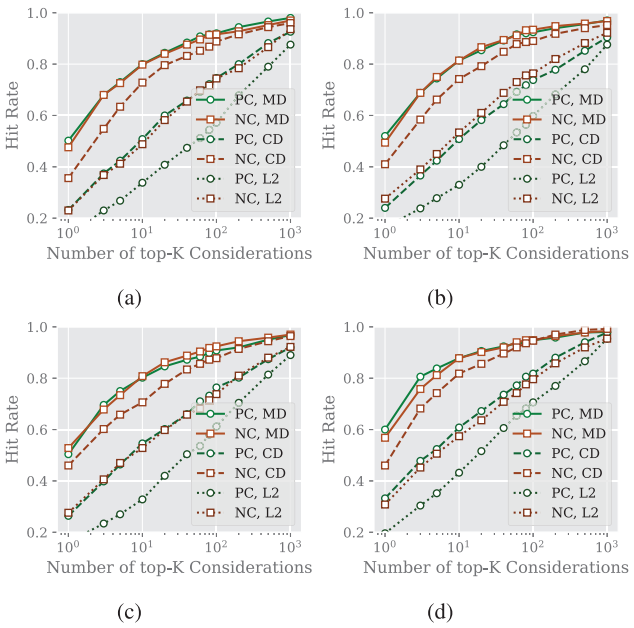


Fig. 16. Probability of including the target material spectrum in the top-K matches (e.g., hit rate) for sample sets with minimum allowable strengths of (a) 0.1, (b) 0.25, (c) 0.5, and (d) 0.75. The six lines correspond to PC versus NC models, and to distribution-based (MD), expectation-based CD, and expectation-based L2 distance (L2) matching schemes.

Appendix D
Randomized-Conditioning Consistency Check

To confirm that the performance improvement of the PC model arises from physically consistent conditioning, an additional consistency check was conducted. Using the same retrieval and matching codes as in Section IV-D, the PropNet and BgNet outputs were randomly permuted across HSI cubes so that each pixel's conditioning vector no longer corresponded to its true atmospheric and background parameters. The resulting hit-rate curves of Fig. 16 indicate that the randomized-conditioning configuration exhibits no measurable gain over the unconditioned baseline for the probabilistic, distribution-based matching scheme, while the deterministic cosine-distance and L2 schemes degrade more substantially. This behavior demonstrates that the model leverages the conditioning information only when it encodes physically meaningful context: the variational MD formulation effectively reverts to baseline performance when the physics is inconsistent, whereas deterministic schemes are more sensitive to mismatched conditioning.

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